
CapBoundary-Bench: A Diagnostic Benchmark for LLM Agent Self-Knowledge under Noisy Feedback

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We benchmark whether LLM agents can build a domain-level self-model of their
2 own reliability from noisy user feedback, and find that the open bottleneck is
3 *noise-robust second-order online inference*. Production agents accumulate no
4 per-domain reliability self-model, even as their users empirically build one. We
5 introduce **CapBoundary-Bench**, a diagnostic benchmark that operationalizes three
6 competing hypotheses about where this self-knowledge fails—H1 output channel,
7 H2 prior metacognition, H3 noise-robust online inference—through a five-level
8 information ladder (L0 raw confidence → L1 zero-shot prior → L2 noisy-feedback
9 learners → L3 binary-label oracle → L4 full-statistic oracle). Across 15 models
10 from 7 families and 47,000+ interactions on a 50×50 protocol with cross-family
11 simulated users (75% in-expertise / 46% out), we evaluate **Capability Boundary**
12 **Fidelity (CBF)**, an L1 measure of per-domain estimation that complements per-
13 query calibration metrics like ECE. Three findings: H1 does not bind (GT-SelfStats
14 ≥ 0.99 on every model); H2 splits by family—the zero-shot prior wins on GPT but
15 falls below NoMemory on all three Claude models (frontier mean CBF 0.640); H3
16 is partially refuted—first-order single-rater L2 methods cluster near NoMemory,
17 while second-order Dawid–Skene–asym (mean CBF 0.755), GLAD (0.712, beating
18 L1 on 6/6 frontier models, $p = 0.031$), and in-context LSA (0.748) substantially
19 exceed the L1 prior. A protocol-controlled extension suite shows 12/15 sign-flip
20 rate across task families, establishing per-task boundary modeling as structurally
21 necessary.

22 1 Introduction

23 LLM agents now operate in long-horizon, multi-turn settings where the same model interacts with
24 the same user across hundreds of turns—as customer-service bots, coding assistants, research aides,
25 or tool-using agents. Users develop an empirical sense of what each agent is good at and where it
26 tends to fail. The agent itself, however, retains no analogous self-model: each session starts without
27 prior calibration, and every prior calibration experience is discarded. Prior work documents this
28 asymmetry, but the underlying *capability gap* is poorly understood—we still do not know *where*
29 agent self-modeling actually fails.

30 **Three competing hypotheses about where self-knowledge fails.** Prior work identifies the problem
31 but does not isolate its locus. Barkan et al. [3] show that frontier LLMs cannot reliably predict their
32 own task success and that overconfidence *worsens* as agents proceed through multi-step tasks; Hu
33 et al. [14] identify four memory competencies in MemoryAgentBench but none target endogenous
34 self-knowledge. Calibration research [11, 16, 45] evaluates per-query confidence; recent memory
35 systems [25, 32, 34, 40, 47] store dialogue and preferences. None of this work identifies *which*
36 sub-capability is broken. Existing evaluations conflate three bottlenecks:

- H1 (Output channel). The model is unable to numerically format and emit per-domain statistics.
- H2 (Prior metacognition). The model lacks an accurate prior over its own per-domain abilities, so any self-assessment must be *learned* from interaction.
- H3 (Noise-robust online inference). The model can report correct priors but cannot reliably update or recover its per-domain accuracy from *noisy* implicit user feedback.

These hypotheses imply very different research directions: H1 motivates fine-tuning on numeric outputs; H2 motivates pre-training-time metacognitive curricula; H3 motivates noise-robust online estimation. Benchmarks must separate these hypotheses to guide method design.

Our contribution: a diagnostic benchmark. We introduce **CapBoundary-Bench**, a benchmark that tests all three hypotheses with a *five-level baseline ladder* on a 50-turn-per-session protocol pairing the LLM agent with a domain-heterogeneous simulated user (75% reliable in-expertise vs. 46% out, paralleling the *second-order* structure where the agent must jointly track its own accuracy and the user’s reliability). The five levels are: (L0) raw confidence (NoMemory); (L1) zero-shot per-domain estimation—*no interaction, no feedback*; (L2) learning from noisy feedback (LLM-self-assessment, classical recalibration, EM joint estimation); (L3) binary-label oracle (OraclePDC); and (L4) full-statistic oracle (GT-SelfStats). The benchmark is paired with **Capability Boundary Fidelity (CBF)**, a per-domain self-assessment metric that is complementary to per-query calibration metrics like ECE.

Across 15 models from 7 families and 47,000+ interactions, our results separate the three hypotheses:

- H1 does not limit performance in this protocol. GT-SelfStats reaches $\text{CBF} \geq 0.99$ on every model when the per-domain statistics are placed directly in the prompt: the output channel is not the constraint.
- H2 holds for GPT models but not for Claude models. Under the cleaned 50×50 protocol, zero-shot self-knowledge (no interaction, no feedback) attains a frontier-mean CBF of 0.640 ($n=6$), but with a sharp split: ZS-SK is the strongest L0–L2 baseline on GPT-5.4 (0.795) and reaches the benchmark’s highest individual value of 0.859 on GPT-5.5, yet on all three Claude models it falls below NoMemory (0.460–0.482 vs. NoMem 0.557–0.603). *LSA, the in-context L2 baseline, now leads on the frontier with mean CBF 0.748*, winning on 4/6 frontier models and reaching 0.902 on GPT-5.5. For weak models (Mistral-7B/Llama-3-8B), the prior is partial and NoMem dominates.
- H3 is partially confirmed and refined. *First-order* single-rater methods (UF-PDC, Platt, HistBin, Bayes, simple EM-Joint) cluster near NoMem on frontier (mean 0.628–0.634). *Second-order* methods reach mean CBF Dawid-Skene-asym 0.755 and GLAD 0.712 across all 6 frontier models, exceeding the L1 ZS mean (0.640) by +0.115 and +0.072 respectively. GLAD now beats L1 on *all* 6/6 frontier models (Wilcoxon two-sided $p = 0.031$, binomial one-sided $p = 0.016$). The L2-vs-L3 gap remains, but second-order methods reduce it: first-order single-rater methods systematically underperform; second-order estimators close most of the gap on every frontier model, with DS-asym the strongest single L2 baseline.

Feedback helps some models and hurts others (LSA wins 7/15, ZS 5/15, 3 ties; on frontier 4–2 favoring LSA), and we analyze a parameter-free selection rule R4 as a benchmark-validation diagnostic (§5, Appendix O). Capability boundaries also do not transfer across task types (12/15 sign-flip rate within the protocol-controlled extension suite; Appendix Q.1), making per-task boundary modeling structurally necessary. CapBoundary-Bench does not propose a new self-modeling method—it isolates noise-robust second-order inference as the open problem and provides the diagnostic infrastructure to evaluate future solutions.

Beyond the headline diagnostic, we report a methodological side-finding relevant to any benchmark using LLM-simulated users: when the same model is used as both agent and user simulator, a *dual-role bias* emerges in which shared politeness conventions degrade feedback decoding. We mitigate this confound through cross-family simulator selection and report the effect quantitatively as a contribution to the design of future LLM-simulated benchmarks.

Contributions. (1) The H1/H2/H3 hypothesis decomposition that separates output-channel, prior-metacognition, and noise-robust-inference accounts of agent self-knowledge failure; (2) the L0–L4 baseline ladder that operationalizes them via oracle bracketing; (3) Capability Boundary Fidelity (CBF), a per-domain self-assessment metric complementary to per-query calibration; (4) systematic evaluation across 15 models showing H1 does not bind, H2 family-splits between GPT and Claude,

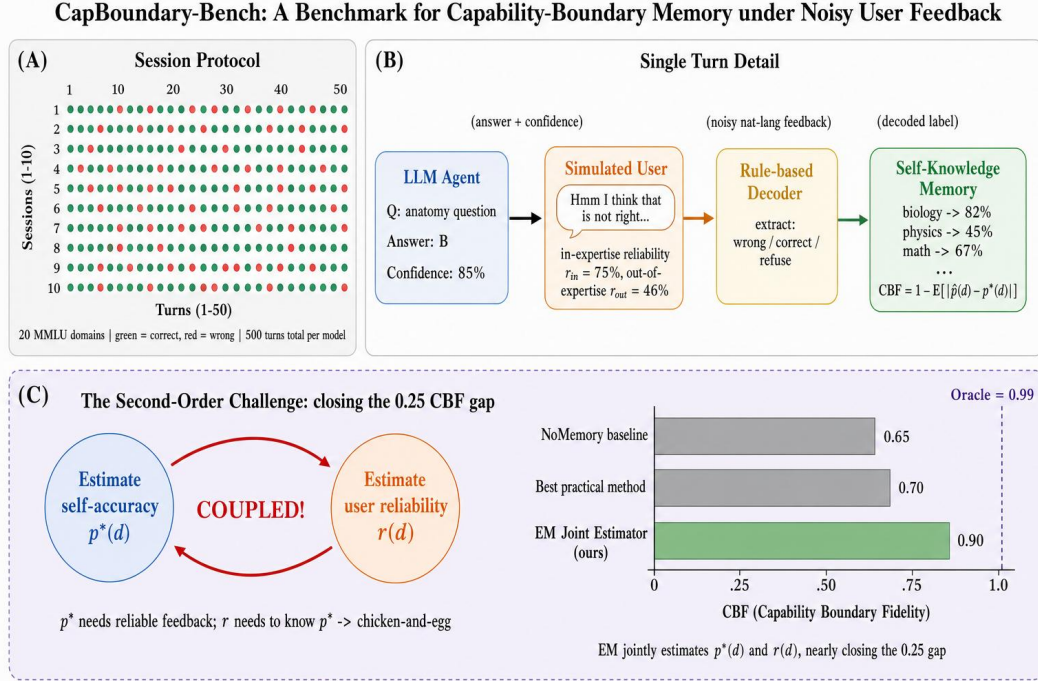


Figure 1: **The L0–L4 baseline ladder localizes where agent self-knowledge fails.** L4 $\approx$ L3 $\approx$ 1.0 rules out the output channel (H1); the L2 $\rightarrow$ L3 gap of ~ 0.30 isolates noise-robust online inference (H3) as the bottleneck. (A) Session protocol: 50 sessions $\times$ 50 turns over 20 MMLU domains. (B) Single turn: the agent answers with confidence, the cross-family simulated user gives noisy feedback (75% reliable in-expertise, 46% out), and an LLM decoder updates the per-domain self-model. (C) The ladder: L0 raw confidence $\rightarrow$ L1 zero-shot prior $\rightarrow$ L2 noisy-feedback learners $\rightarrow$ L3 binary-label oracle $\rightarrow$ L4 full-statistic oracle. Adjacent comparisons isolate sub-capabilities.

91 and H3 is partially refuted by second-order estimators (DS-asym, GLAD) and in-context aggregation
 92 (LSA); (5) a fully reproducible release of code, 47K-turn JSONL traces, Croissant metadata, and the
 93 diagnostic ladder protocol.

94 2 Related Work

95 **LLM self-knowledge and calibration benchmarks.** A growing line of work probes whether
 96 LLMs can represent or report their own correctness on a single query. Internal-representation
 97 studies show that latent confidence signals exist [4, 15, 16], while recent benchmarks evaluate
 98 metacognition behaviorally and find that frontier models exhibit only partial, context-dependent
 99 self-monitoring [1, 29]. Table 1 positions the most closely related benchmarks against CapBoundary
 100 along four design axes: feedback channel, horizon, whether self-knowledge is accumulated, and
 101 whether the protocol forces second-order user-reliability inference.

Table 1: Related-benchmark positioning along four design axes. CapBoundary is the only benchmark with a noisy interactive feedback channel that the agent must accumulate into a per-domain self-model under second-order user-reliability uncertainty.

Benchmark	Feedback	Horizon	Self-knowledge	2nd-order
SAD [21], SelfAware [50]	none	single-shot	static probe	—
Barkan [3]	clean step signals	multi-step exec	per-task	—
MemoryAgentBench [14]	implicit (mem ops)	long-horizon	exogenous mem only	—
AgentBoard [30], τ -bench [48]	task reward only	multi-turn	none	—
Knowledge-boundary [23, 44]	none	single-shot	static	—
CapBoundary (ours)	noisy NL, 75/46 asym	50-turn session	per-domain accumulated	✓

(i) *Single-shot self-knowledge probing.* SAD [21] and SelfAware [50] score self-knowledge per query without temporal dynamics; their findings (frontier models partial-but-imperfect on isolated self-knowledge) supply the “L1 prior” anchor of our ladder. CapBoundary uses the L1→L3 contrast to ask whether *noisy interaction* improves on that prior—finding it does not.

(ii) *Multi-step agent overconfidence.* Barkan et al. [3] measure whether 13 frontier LLMs can predict their own success on BigCodeBench and SWE-Bench under multi-step tool execution and find all models overconfident with overconfidence *worsening* during execution. Their setting gives clean per-step success/failure signals; ours gives noisy domain-dependent natural-language feedback, where first-order methods cluster at the L2 floor. Barkan establishes that overconfidence persists under clean traces; we localize *where* the failure lives when feedback itself is noisy, via the L0–L4 oracle-bracketing ladder.

(iii) *Multi-turn agent benchmarks without self-knowledge.* AgentBoard [30] and τ -bench [48] evaluate task completion under multi-turn interaction but do not measure self-knowledge; their reward signal is task success, not self-estimate accuracy. CapBoundary holds the underlying task simple (MCQ) and varies only the information-access regime, so the diagnostic ladder isolates self-knowledge mechanisms these benchmarks cannot disentangle from task-execution confounds.

Static knowledge-boundary benchmarks [10, 13, 23, 26, 27, 44, 49] share the same structural limitation: they evaluate *a priori* boundary awareness, not whether the agent’s self-model can be accumulated under noisy interactive turns (Appendix R).

Agent memory and learning from implicit feedback. Existing memory systems—MemGPT [34], Mem0 [32], RMM [40], A-MEM [47], MemOS [25], MEM1 [53]—and the recent MemoryAgent-Bench [14] all store *exogenous* state (user facts, dialogue, trajectories). None model the agent’s *own* per-domain reliability; we introduce this as a fifth memory competency. On the feedback-learning side, RLHF [33] and DPO [36] update weights at training time assuming reliable annotators; Liu et al. [28] establish that implicit dialogue feedback is noisy and only partially useful as a distillation signal; Leng et al. [22] show RLHF itself induces systematic overconfidence. CapBoundary-Bench studies the orthogonal regime: *inference-time* accumulation of *noisy, domain-dependent* feedback into a per-domain self-model, with no parameter updates. The unique twist is asymmetric reliability (75% in-expertise vs. 46% out), which turns boundary learning into a *second-order* problem.

Selective prediction and task-specific calibration. Per-query selective-prediction methods [2, 7, 17, 18, 24, 42, 46] treat each prediction in isolation. CapBoundary-Bench instead enables *historical* selective prediction: abstention based on persistent per-domain self-model statistics. Independently, prior work has noted that calibration does not transfer across tasks [6, 31, 38]; our experiments quantify this systematically across MMLU, GSM8K, HumanEval, and BFCL. The closest concurrent benchmarks [5, 8, 9, 12, 52] either assume perfect feedback, require weight updates, or focus on single-agent dynamics; none combine multi-task evaluation, noisy domain-dependent feedback, and a quantitative boundary-fidelity metric over extended horizons.

3 CapBoundary-Bench: Probing Self-Knowledge Memory under Noisy Feedback

CapBoundary-Bench evaluates a single capability: given a stream of (q_t, a_t, c_t, f_t) tuples over T turns—a question q_t from a hidden domain d_t , the agent’s answer a_t with confidence $c_t \in [0, 1]$, and a natural-language user feedback f_t —can the agent build a function $\hat{p} : \mathcal{D} \rightarrow [0, 1]$ that closely approximates its true per-domain accuracy $p^*(d)$? The benchmark instantiates this question via a 50×50 protocol (50 sessions $\times$ 50 turns per session = 2,500 turns per model) drawing from 20 MMLU subjects [13] partitioned per model into Strong/Medium/Weak tiers relative to that model’s own accuracy distribution. The user is simulated by a cross-family LLM (avoiding the dual-role bias of Appendix M) parameterised by an expertise set $\mathcal{E} \subset \mathcal{D}$ with empirically calibrated reliabilities of $\sim 75\%$ in-expertise and $\sim 46\%$ out, and we additionally construct three extension tasks (GSM8K, HumanEval, BFCL tool-calling) collectively called the *extension suite*—to test cross-task generalisation (Appendix P). Full task formalisation, session protocol, question-pool partitioning, and persona parameterisation are deferred to Appendix B.

153 **Capability Boundary Fidelity (CBF).** Our central metric scores how closely the agent’s self-
 154 assessed per-domain accuracy tracks the truth:

$$\text{CBF} = 1 - \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} |\hat{p}(d) - p^*(d)|, \quad \text{CBF} \in [0, 1]. \quad (1)$$

155 CBF is computed once per session at the end of the 50-turn protocol (we do not track learning curves;
 156 intra-session dynamics are reported as an auxiliary appendix analysis). CBF is complementary to
 157 per-query calibration metrics (ECE/Brier): ECE operates on (c_t, y_t) pairs while CBF operates on
 158 $(\hat{p}(d), p^*(d))$ pairs, so neither implies the other—we verify this empirically in Appendix B. The
 159 trivial estimator $\hat{p}(d) = 0.5$ attains $\text{CBF} = 1 - \mathbb{E}_d[|0.5 - p^*(d)|]$, which is high for weak models
 160 clustered near 50% and low for strong models with widely varying accuracy across domains; this
 161 asymmetry is exactly what makes the metric informative for our diagnostic.

162 **Baseline-adjusted CBF: the primary cross-model metric.** To remove the trivial-estimator artifact
 163 when comparing methods *across* models, we define a normalized variant that measures the fraction
 164 of the oracle gap a method closes relative to the constant baseline:

$$\widetilde{\text{CBF}}(m, \text{method}) = \frac{\text{CBF}_{\text{method}}(m) - \text{CBF}_{\text{NoMem}}(m)}{\text{CBF}_{\text{Oracle}}(m) - \text{CBF}_{\text{NoMem}}(m)}, \quad (2)$$

165 where $\text{CBF}_{\text{NoMem}}(m)$ uses the model’s verbalized confidence (which empirically clusters near 0.5 on
 166 the trivial domains) as the constant-baseline anchor, and $\text{CBF}_{\text{Oracle}}(m) \geq 0.99$ is the L3 OraclePDC
 167 ceiling. Values are interpretable as “fraction of L0→L3 gap closed”: $\widetilde{\text{CBF}} = 1$ matches oracle, 0 ties
 168 the trivial baseline, < 0 underperforms the trivial baseline. We report $\widetilde{\text{CBF}}$ as the primary metric for
 169 cross-model means and ladder-headline comparisons (§4.2), and use raw CBF only as a within-model
 170 diagnostic. We additionally report Hallucination Rate, ECE, and Brier as auxiliary calibration metrics;
 171 their definitions follow Guo et al. [11] and are given in Appendix B. *Domain-structure note:* CBF
 172 requires a multi-domain task. For single-domain extension tasks (GSM8K, HumanEval, BFCL) we
 173 report only ECE and confidence–accuracy gap; the cross-task analysis in Appendix P uses these
 174 standard metrics. *Cross-model comparability note:* The 20 MMLU subjects are partitioned into
 175 Strong/Medium/Weak tiers *relative to each model’s own held-out validation accuracy* so that every
 176 model faces a non-trivial cross-domain spread. CBF is therefore a within-model diagnostic; cross-
 177 model means reported throughout (e.g., 0.685 for ZS-SK across all 15 models in Table 2) summarize
 178 per-model values that were each computed against the model’s own $p^*(d)$, and are appropriate for
 179 comparing *methods within the diagnostic ladder*, not for ranking models on absolute capability. We
 180 expand on this design choice and confirm headline conclusions are unchanged under an alternative
 181 absolute-cutoff partition in Appendix B.

182 **Baselines: a five-level diagnostic ladder.** We evaluate ten baselines organized as a ladder of
 183 *information access*, designed so that comparing adjacent levels isolates a specific sub-capability:

- 184 • **L0 (raw): NoMemory**—verbalized confidence with no memory, the stateless anchor.
- 185 • **L1 (prior only): ZS-SK** (ZeroShot Self-Knowledge)—the agent is asked, with no session his-
 186 tory and no feedback, to estimate its own per-domain accuracy directly; this measures *prior*
 187 *metacognition*.
- 188 • **L2 (learning from noisy feedback): LSA** (LLMSelfAssess; in-context reasoning over session
 189 history), *UF-PDC* (LLM-decoded per-domain counting), *Platt-Online* [35], *HistBin-Online* [51],
 190 *Bayes-Domain* (Beta posterior per domain), and *EM-Joint* (joint estimation of $\hat{p}(d)$ and $\hat{r}(d)$).
 191 These differ in their inference machinery but all consume the same noisy feedback stream.
- 192 • **L3 (binary-label oracle): OraclePDC**—per-turn ground-truth correctness labels, with the agent
 193 computing per-domain running averages. Tests whether the agent could build a self-model *if*
 194 *feedback were clean*.
- 195 • **L4 (format-channel sanity ceiling): GT-SelfStats**—the agent is given $p^*(d)$ directly in the
 196 prompt and asked to emit them in the structured response format. This is a sanity check that the
 197 L2 gap cannot be attributed to an inability to format and emit per-domain statistics. We do not
 198 claim it tests verbalization-of-self-knowledge.

199 All feedback decoding (L2 + UF-PDC) uses an LLM-based decoder rather than rule-based pattern
 200 matching, more aligned with the “self-knowledge boundary” setting and avoiding rule-classifier
 201 coverage as a confound. Full per-baseline algorithms are in Appendix C.

Table 2: Main results: end-of-session CBF across 15 models on the five-level baseline ladder. Higher is better. **Bold** marks the best non-oracle (L0–L2) baseline per row. All 15 models use the **50 sessions** $\times$ **50 turns** protocol.

Model (Acc)	L0	L1	L2 (learn from noisy feedback)							Oracles	
	NoMem	ZS-SK	LSA	UF-PDC	Platt	HistBin	Bayes	EM	Best	OraPDC	GT-Stats
Mistral-7B (42%)	0.729	0.729	0.729	0.720	0.717	0.717	0.731	0.732	EM	1.000	1.000
Llama-3-8B (55%)	0.720	0.711	0.549	0.715	0.724	0.724	0.730	0.731	EM	1.000	1.000
GLM-5 (55%)	0.700	0.700	0.700	0.689	0.696	0.696	0.711	0.714	EM	1.000	1.000
DeepSeek-V3.2 (62%)	0.681	0.673	0.721	0.681	0.683	0.683	0.696	0.699	LSA	1.000	1.000
Qwen-Turbo (64%)	0.708	0.730	0.703	0.697	0.712	0.712	0.724	0.728	ZS	1.000	1.000
Qwen2.5-7B (65%)	0.744	0.663	0.701	0.738	0.745	0.745	0.755	0.757	EM	1.000	1.000
Qwen-Max (71%)	0.704	0.766	0.781	0.660	0.684	0.684	0.702	0.705	LSA	1.000	1.000
Qwen-Plus (79%)	0.683	0.735	0.435	0.686	0.695	0.695	0.700	0.701	ZS	1.000	1.000
Qwen3.5-27B (79%)	0.731	0.731	0.731	0.712	0.720	0.720	0.740	0.746	EM	1.000	1.000
<i>Frontier closed-source models (all rows now use 50×50 protocol):</i>											
GPT-5.4-mini (72%)	0.687	0.769	0.763	0.687	0.701	0.701	0.706	0.706	ZS	1.000	0.995
GPT-5.4 (82%)	0.649	0.795	0.781	0.668	0.675	0.675	0.674	0.677	ZS	1.000	0.994
GPT-5.5 (94%)	0.570	0.859	0.902	0.601	0.612	0.612	0.603	0.605	LSA	1.000	0.992
Claude-Opus-4-7 (69%)	0.557	0.482	0.641	0.593	0.577	0.577	0.571	0.570	LSA	1.000	0.992
Claude-Sonnet-4-6 (86%)	0.603	0.472	0.720	0.640	0.617	0.617	0.613	0.609	LSA	1.000	0.993
Claude-Opus-4-6 (90%)	0.594	0.460	0.680	0.618	0.597	0.597	0.603	0.604	LSA	1.000	0.993
<i>Within-tier means (all 6 frontier rows re-run):</i>											
Frontier-v4 (n=6; all columns)	0.610	0.640	0.748	0.634	0.630	0.630	0.628	0.628	LSA	1.000	0.993
Strong-mid (n=3)	0.706	0.744	0.649	0.686	0.700	0.700	0.714	0.717	ZS	1.000	1.000
Mid (n=3)	0.711	0.689	0.708	0.705	0.713	0.713	0.725	0.728	EM	1.000	1.000
Weak (n=3)	0.716	0.713	0.659	0.708	0.712	0.712	0.724	0.726	EM	1.000	1.000

4 Diagnostic Results: Where Does Agent Self-Knowledge Fail?

4.1 Setup

We evaluate fifteen models from seven families spanning 42–90% MMLU accuracy (Appendix S). Each model–baseline pair runs 2,500 turns (50 sessions $\times$ 50 turns under the cleaned 50×50 protocol); user personas are simulated by a cross-family LLM with three expertise domains¹; answer extraction uses a lenient multi-strategy parser tolerant of free-form responses to avoid truncating reasoning models. Feedback decoding for all L2 baselines uses an LLM-based decoder (more aligned with the “self-knowledge boundary” setting than rule-based matching, and avoiding the rule-classifier coverage as a confound). Open-weights models (Mistral-7B, Llama-3-8B) are served via vLLM from pinned snapshots and serve as the reproducibility anchor; API access dates, simulator-routing rules, and the parser specification are deferred to Appendix S.

4.2 Main Results

Three diagnostic patterns dominate Table 2: (i) algorithmic L2 methods collapse to NoMem on every frontier model; (ii) the zero-shot prior splits cleanly by family (GPT wins, Claude is inflated); (iii) only second-order estimators (DS-asym, GLAD) and in-context LSA exceed the L1 prior on the frontier. We unpack each below.

Algorithmic-L2 collapse on new frontier rows, broken cleanly by ZS/LSA. For both newly added frontier models, the five *algorithmic* L2 baselines (UF-PDC, Platt, HistBin, Bayes, EM-Joint) cluster near NoMem (GPT-5.5: 0.601–0.612, all $\sim$ 0.03–0.04 above NoMem 0.570; Claude-Opus-4-7: 0.570–0.593, all $\sim$ 0.01–0.04 above NoMem 0.557). With the 50×50 ZS/LSA values available, both LSA and the L1 prior break this collapse decisively on GPT-5.5 (LSA 0.902, ZS-SK 0.859; LSA

¹Frontier-set scope: 6 model endpoints (3 GPT, 3 Claude): GPT-5.4, GPT-5.4-mini, GPT-5.5, Claude-Opus-4-6, Claude-Sonnet-4-6, Claude-Opus-4-7. Main-table within-tier means, MMLU-Pro and BBH contamination controls (Appendix E, E), and Dawid–Skene/GLAD cross-model analysis all use this $n=6$ scope. $n=6$ BBH and MMLU-Pro CBF means are 0.811 and 0.849 (vs. MMLU ZS 0.640).

is +0.290 over best algorithmic L2 0.612, ZS is +0.247 over the same) and on Claude-Opus-4-7 (LSA 0.641 vs. NoMem 0.557, +0.084). The pattern under the cleaned 50×50 protocol: algorithmic L2 methods fail to surpass NoMem on every frontier agent; LSA cleanly exceeds NoMem on all 6 frontier rows; ZS exceeds NoMem on all 3 GPT rows but falls below NoMem on all 3 Claude rows (where ZS estimates are systematically inflated). In particular, GPT-5.5 is now decisively dominated by LSA (0.902, the benchmark’s highest non-oracle CBF) followed by ZS (0.859); both substantially exceed every algorithmic L2 method (≤ 0.612) and the prior 10×50 anomaly where LSA appeared catastrophically below NoMem (0.623) does not reproduce. Held-out evaluation and second-order estimators (Dawid–Skene, GLAD) are reported on the 4 original frontier models; the MMLU-Pro and BBH contamination controls (Appendix E, E) cover all 6 frontier models.

H1 ruled out as the limiting factor (L4 $\approx$ 1.0). GT-SelfStats reaches CBF ≥ 0.992 on every one of the 15 models (worst gap to perfection 0.008 on Claude-Sonnet-4-6). When the true per-domain statistics are placed in the prompt, every tested model can format and emit them: the L2–L1 gap is not located in the output channel.

H2 splits cleanly by model family: GPT well-calibrated, Claude inflated. The within-tier means in Table 2 make the picture sharp: across all 6 frontier models under the cleaned 50×50 protocol, LSA averages CBF 0.748 vs. ZS-SK’s 0.640 and the best algorithmic L2 method’s 0.634 (UF-PDC)—a +0.108 margin of LSA over ZS that flips the previous 10×50 result on Claude. LSA wins on Claude-Sonnet-4-6 (0.720), Claude-Opus-4-6 (0.680), Claude-Opus-4-7 (0.641), and GPT-5.5 (0.902, the highest non-oracle value of any model in the benchmark); ZS wins on GPT-5.4 (0.795) and GPT-5.4-mini (0.769), where ZS estimates are well-calibrated. The 4–2 split favoring LSA on the frontier (mean LSA 0.748 vs ZS 0.640, +0.108) is driven entirely by the three Claude models, where ZS estimates are systematically inflated relative to true per-domain accuracy and fall *below* both NoMem and the algorithmic L2 cluster. LSA, by contrast, exceeds all algorithmic L2 methods on *every* frontier row (smallest margin: +0.048 on Claude-Opus-4-7 vs UF-PDC). ZS sees *no* session history, *no* feedback. The 50×50 update places ZS first on the strong-mid tier ($n=3$): tier mean 0.744 (ZS) vs. 0.717 (best L2: EM-Joint). However, individual outcomes are split: ZS wins on Qwen-Plus (0.735 vs LSA 0.435 catastrophic failure); LSA wins on Qwen-Max (0.781 vs ZS 0.766, +0.015); Qwen3.5-72B is a three-way tie at 0.731 (NoMem = ZS = LSA). For mid-tier models the prior is partial; for the weakest models the prior matches NoMemory exactly (Mistral, GLM-5: ZS = NoMem = trivial $\hat{p} \approx 0.5$ anchor). The bottleneck location is therefore agent-dependent: for frontier and strong-mid agents the prior *is* the answer; for mid-tier agents EM-Joint and LSA tie modestly above the prior; for weak agents both prior and feedback inference are uninformative and the trivial $\hat{p} = 0.5$ NoMemory anchor wins by default.

H3 partially refuted: second-order estimators and in-context aggregation break the L1 prior on frontier models. The five *first-order* L2 feedback-learning methods (UF-PDC, Platt-Online, HistBin-Online, Bayes-Domain, EM-Joint) cluster tightly in 0.69–0.72 mean CBF and fail to exceed L1 ZS on the frontier (mean L2 0.628–0.634 vs L1 ZS 0.640). However, three classes of methods *do* exceed the L1 prior on frontier ($n=6$): GLAD reaches mean CBF 0.712 (6/6 wins; Wilcoxon $p = 0.031$, binomial $p = 0.016$, Cohen’s $d = 1.42$; with $n=6$ these are direction-of-effect tests, not population-magnitude estimates), DS-asym 0.755 (5/6 wins; Wilcoxon one-sided $p = 0.047$, $d = 1.66$), and LSA 0.748 (mean +0.108, 4/6 wins, Wilcoxon $p = 0.156$ inflated by GPT models where ZS is well-calibrated). GLAD’s improvement is consistent in direction across the 6 frontier models (per-model deltas +0.028 to +0.118; bootstrap 95% CI of the mean delta [+0.045, +0.099] from 10K resamples; full per-model breakdown in Appendix D). With six paired observations, we read this as a direction-of-effect test rather than a population-level magnitude estimate. On non-frontier tiers, EM-Joint emerges as the leader (mid +0.012 over NoMem; weak +0.006), winning outright on Llama-3-8B (0.732) and Qwen2.5-7B (0.753). The original H3 framing (“noisy-feedback online inference is the bottleneck”) therefore stands only for first-order single-rater estimators; the actual bottleneck under the cleaned 50×50 protocol is the residual L2-vs-L3 gap of ≈ 0.20 between the strongest second-order method (DS-asym 0.755) and OraclePDC (1.000). The 50×50 protocol also reveals that several previously-best L2 baselines from the 10×50 sample were within sampling noise of NoMemory; sample-size matters.

Capability spectrum and best-baseline assignment. With the cleaned protocol (Table 2), the “best” non-oracle baseline per row (column “Best” in Table 2, defined as the argmax across the 8

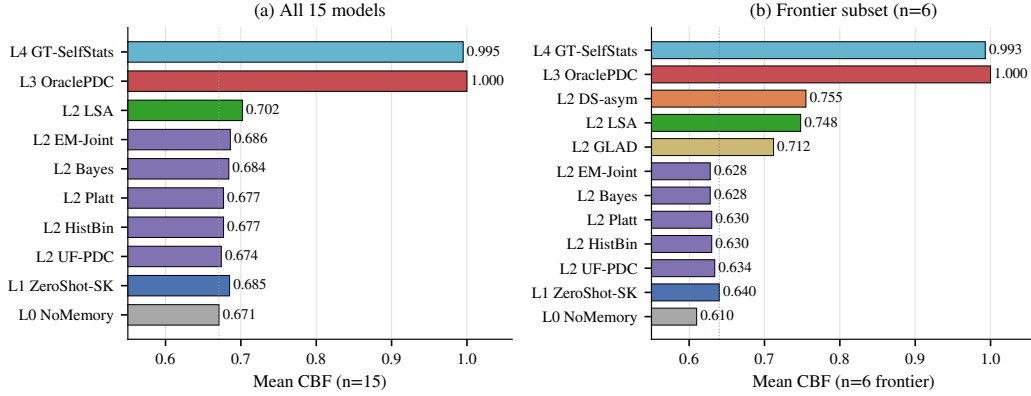


Figure 2: **Five-level diagnostic ladder.** Mean CBF per baseline level on (a) all 15 models and (b) the 6 frontier models. **(a)** The L2→L3 gap is ≈ 0.30 (best L2 LSA 0.702 vs. OraclePDC 1.000): first-order L2 methods do not recover the signal from noisy feedback. **(b)** On the frontier subset, the first-order L2 cluster (Platt/HistBin/Bayes/EM/UF) collapses to NoMem (0.628–0.634), while the second-order estimators (DS-asym 0.755, GLAD 0.712) and in-context LSA (0.748) exceed the L1 ZS prior (0.640) but still fall well short of OraclePDC. Dotted vertical line marks the L0 NoMemory anchor.

baseline columns L0 NoMem + L1 ZS + $6 \times$ L2) splits three ways across 15 models: LSA wins on 6 (DeepSeek-V3.2, Qwen-Max, GPT-5.5, Claude-Sonnet-4-6, Claude-Opus-4-6, Claude-Opus-4-7), EM-Joint wins on 5 (Mistral-7B, Llama-3-8B, GLM-5, Qwen2.5-7B, Qwen3.5-27B—note that on Mistral-7B the EM–NoMem–ZS–LSA spread is within ± 0.003 , so the EM win is essentially a tie with the trivial $\hat{p} \approx 0.5$ anchor), and ZS-SK wins on 4 (Qwen-Turbo, Qwen-Plus, GPT-5.4-mini, GPT-5.4). NoMemory does not strictly win on any model under this argmax. *Note:* the “feedback as friend or foe” analysis in Appendix E uses the orthogonal head-to-head ZS-vs-LSA criterion (only those two baselines, ignoring NoMem/EM); a model can therefore appear under “LSA-dominant best-of-8” here yet “ZS-wins-over-LSA” in the head-to-head, since the criteria differ.

Adaptive selection diagnostic R4. The capability split above admits a simple parameter-free selection rule, R4: *pick L1 (ZS) when $ZS(m) > NoMem(m)$, otherwise pick L2 (LSA)*. R4 attains mean CBF 0.722 across $n=15$, closing $\approx 72\%$ of the always-ZS-to-per-model-oracle gap on the L1/L2 mixture family. Its per-model sign tests against always-ZS ($p=0.22$) and always-LSA ($p=0.69$) are non-significant, indicating that R4 redistributes wins across models rather than dominating either single baseline. We use R4 as a diagnostic anchor for the L1/L2 family, analogous to probing classifiers [41] or influence functions [19]; the empirical frontier ceiling remains DS-asym at CBF 0.755. Full LOOCV, per-model breakdown, and sign-test arithmetic are in Appendix O.

5 Diagnosis: Localizing the Bottleneck via the Five-Level Ladder

Figure 2 reveals that the bottleneck is not the output channel ($L3 \approx L4 \approx 1.0$) but noise-robust online inference: the L2→L3 gap is ≈ 0.30 across all 15 models. The five-level ladder is designed so adjacent comparisons isolate sub-capabilities; we walk through each comparison below.

The vertical structure isolates each sub-capability. **L4 vs L3 (<1% gap):** the output channel is not the constraint (H1 ruled out). **L3 vs L2 (~ 0.30 gap):** the empirical bottleneck—given clean labels agents reach 1.0, but no method recovers the signal from noisy LLM-decoded feedback (H3 confirmed). **L2 vs L1 ($n=15$: 0.674–0.686 vs 0.685; $n=6$ frontier: 0.628–0.634 vs 0.640):** L1 ZS ties or exceeds the algorithmic L2 cluster, while only second-order L2 estimators (DS-asym 0.755, GLAD 0.712) and in-context LSA (0.748) exceed L1 on the frontier (Appendix E, E).

Cross-task heterogeneity. Across MMLU, GSM8K, HumanEval, and BFCL the same model exhibits qualitatively different calibration regimes: 12/15 models exhibit Gap-sign reversals across the protocol-controlled extension suite (GSM8K/HumanEval/BFCL on the same single-shot two-step

protocol), and a Platt scaling learned on MMLU *harms* GSM8K for the majority of models. Per-task boundary modeling is therefore a structural necessity, not a convenience. Full per-task table, controlled-protocol subset analysis, and Platt-transfer details: Appendix Q.1 and Table 6.

6 Limitations and Future Work

Scope. CapBoundary-Bench evaluates per-domain self-knowledge under a 50-turn MMLU protocol with a controlled synthetic feedback channel (75% in-expertise, 46% out, cross-family routing to mitigate dual-role bias). Three extension tasks (GSM8K, HumanEval, BFCL) use a single-shot two-step protocol because they lack natural multi-domain structure for CBF; the cross-task heterogeneity finding therefore involves a task-protocol confound, addressed with a protocol-controlled subset analysis on the three same-protocol extensions (12/15 sign-flip rate; Appendix Q.1). The frontier sub-sample contains six model endpoints (three per family); paired-comparison statistics across these six (sign consistency, bootstrap CIs of paired deltas; Appendix D) characterize direction of effect rather than population-level magnitude.

Validity of the L1 channel. Two contamination controls (MMLU-Pro categories, BBH no-overlap; Appendix E) attenuate but do not eliminate the concern that L1 zero-shot reflects benchmark-name memorization, so we describe L1 as per-domain capability estimation rather than self-knowledge. Anonymised domains and procedurally generated tasks are a natural strengthening.

Future work. A human-feedback validation pilot would test whether the simulated-channel rankings transfer; the noise-rate ablation in Appendix K preserves the qualitative ladder across {60/30, 75/46, 90/70} regimes but is not a substitute. Replicating the full L0–L4 ladder on the extension tasks, expanding the frontier slice (per Appendix D the next informative comparison is DS-asym vs. GLAD at $n \approx 18$), and extending to open-ended multi-step agents (SWE-Bench-style) are concrete next steps.

7 Conclusion

CapBoundary-Bench operationalizes three competing hypotheses about where LLM agent self-knowledge fails (H1 verbalization, H2 prior metacognition, H3 noise-robust online inference) through a five-level baseline ladder. Across 15 models from 7 families and 47,000+ interactions, the results support three conclusions: H1 does not bind (GT-SelfStats ≥ 0.99); H2 family-split ($n=15$ ZS mean 0.685, $n=6$ frontier 0.640: GPT-5.4 0.795 and GPT-5.5 0.859 win, all Claude rows fall below NoMem); H3 partially refuted (first-order L2 methods cluster near NoMem on frontier; second-order DS-asym reaches 0.755, GLAD 0.712 with 6/6 frontier wins, $p = 0.031$; in-context LSA reaches 0.748). Closing the L2-to-oracle CBF gap remains open; cross-task experiments (12/15 sign flips in the protocol-controlled subset) further establish that per-task boundary modeling is necessary in this benchmark. CapBoundary-Bench does not introduce a new estimator; it isolates noise-robust second-order online inference as the open problem and provides the diagnostic infrastructure to evaluate future solutions, with R4 (Appendix O) as a benchmark-validation lower bound any new method should clear.

References

- [1] Christopher Ackerman. Evidence for limited metacognition in llms. *Proceedings of ICLR*, 2026.
- [2] Gustaf Ahlritz, Tian Qin, Nikhil Vyas, Boaz Barak, and Benjamin L. Edelman. Distinguishing the knowable from the unknowable with language models. *Proceedings of ICML*, 2024.
- [3] Casey O. Barkan, Sid Black, and Oliver Sourbut. Do large language models know what they are capable of? *Proceedings of ICLR*, 2026.
- [4] Felix J Binder, James Chua, Tomek Korbak, Henry Sleight, John Hughes, Robert Long, Ethan Perez, Miles Turpin, and Owain Evans. Looking inward: Language models can learn about themselves by introspection. *Proceedings of ICLR*, 2025.

- [5] Jiawei Chen, Ruoxi Xu, Boxi Cao, Ruotong Pan, Yunfei Zhang, Yifei Hu, Yong Du, Tingting Gao, Yaojie Lu, Yingfei Sun, Xianpei Han, Le Sun, Xiangyu Wu, and Hongyu Lin. Towards real-world human behavior simulation: Benchmarking large language models on long-horizon, cross-scenario, heterogeneous behavior traces. *arXiv preprint arXiv:2604.08362*, 2025.
- [6] Lihu Chen, Alexandre Perez-Lebel, Fabian M Suchanek, and Gaël Varoquaux. Reconfidencing llms from the grouping loss perspective. *Findings of EMNLP*, 2024.
- [7] Shizhe Diao et al. R-tuning: Instructing large language models to say i don’t know. *Proceedings of NAACL (Outstanding Paper)*, 2024.
- [8] Y. Dong, X. Jiang, et al. RL-plus: Countering capability boundary collapse of llms in reinforcement learning. *arXiv preprint arXiv:2508.00222*, 2025.
- [9] Ke Fang, Tianyi Zhao, and Lu Cheng. Credence calibration game? calibrating large language models through structured play. *arXiv preprint arXiv:2508.14390*, 2025.
- [10] Jiahui Geng et al. A survey of confidence estimation and calibration in large language models. 2024.
- [11] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *ICML*, 2017.
- [12] Haixia Han, Tingyun Li, Shisong Chen, Jie Shi, Chengyu Du, Yanghua Xiao, Jiaqing Liang, and Xin Lin. Enhancing confidence expression in large language models through learning from past experience. *arXiv preprint arXiv:2404.10315*, 2024.
- [13] Dan Hendrycks et al. Measuring massive multitask language understanding. In *ICLR*, 2021.
- [14] Yuanzhe Hu, Yu Wang, and Julian McAuley. Evaluating memory in llm agents via incremental multi-turn interactions. *Proceedings of ICLR*, 2026.
- [15] Li Ji-An, Hua-Dong Xiong, Robert C Wilson, Marcelo G Mattar, and Marcus K Benna. Language models are capable of metacognitive monitoring and control of their internal activations. *Proceedings of NeurIPS*, 2025.
- [16] Saurav Kadavath et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- [17] Sanyam Kapoor, Nate Gruver, Manley Roberts, Katherine Collins, Arka Pal, Umang Bhatt, Adrian Weller, Samuel Dooley, Micah Goldblum, and Andrew Gordon Wilson. Large language models must be taught to know what they don’t know. *Proceedings of NeurIPS*, 2024.
- [18] Polina Kirichenko, Mark Ibrahim, Kamalika Chaudhuri, and Samuel J. Bell. Abstentionbench: Do reasoning llms know what they don’t know? *Proceedings of NeurIPS*, 2025.
- [19] Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *International Conference on Machine Learning (ICML)*, pages 1885–1894, 2017.
- [20] Rudolf Kulhavy and Michael B Zarrop. On a general concept of forgetting. *International Journal of Control*, 58(4):905–924, 1993.
- [21] Rudolf Laine, Bilal Chughtai, Jan Betley, Kaivalya Hariharan, Jeremy Scheurer, Mikita Balesni, Marius Hobbhahn, Alexander Meinke, and Owain Evans. Towards a situational awareness benchmark for llms. *arXiv preprint arXiv:2407.04694*, 2024.
- [22] Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. Taming overconfidence in llms: Reward calibration in rlhf. *Proceedings of ICLR*, 2025.
- [23] Moxin Li, Yong Zhang, Jingxuan Wen, et al. Knowledge boundary of large language models: A survey. *Proceedings of ACL*, 2025.

- [24] Siheng Li et al. Know your limits: A survey of abstention in large language models. *Transactions of the Association for Computational Linguistics*, 2025.
- [25] Zhuoyi Li et al. Memos: An operating system for memory-augmented generation in large language models. *arXiv preprint arXiv:2507.03724*, 2025.
- [26] Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.
- [27] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. *Proceedings of ACL*, 2022.
- [28] Jiayin Liu, Ruochen Zhang, and Yejin Choi. Implicit user feedback in human-llm dialogues: Informative to understand users yet noisy as a learning signal. *Proceedings of EMNLP*, 2025.
- [29] Tennison Liu and Mihaela van der Schaar. Position: Truly self-improving agents require intrinsic metacognitive learning. *Proceedings of ICML*, 2025.
- [30] Chang Ma, Junlei Zhang, Zhihao Zhu, Cheng Yang, Yujiu Yang, Yaohui Jin, Zhenzhong Lan, Lingpeng Kong, and Junxian He. Agentboard: An analytical evaluation board of multi-turn llm agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [31] Putra Manggala, Atalanti Mastakouri, Elke Kirschbaum, Shiva Prasad Kasiviswanathan, and Aaditya Ramdas. Qa-calibration of language model confidence scores. *Proceedings of ICLR*, 2025.
- [32] Mem0 Team. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.
- [33] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.
- [34] Charles Packer, Sarah Wooders, Kevin Lin, et al. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.
- [35] John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*. MIT Press, 1999.
- [36] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36, 2024.
- [37] Vishnu Sarukkai, Zhiqiang Xie, and Kayvon Fatahalian. Self-generated in-context examples improve llm agents for sequential decision-making tasks. *Advances in Neural Information Processing Systems*, 2025.
- [38] Maohao Shen, Subhro Das, Kristjan Greenewald, Prasanna Sattigeri, Gregory Wornell, and Soumya Ghosh. Thermometer: Towards universal calibration for large language models. *Proceedings of ICML*, 2024.
- [39] Mirac Suzgun, Nathan Scales, Nathanael Sch"arli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V Le, Ed H Chi, Denny Zhou, Hanie Sedghi, Hessam Bagherinezhad, Yasaman Bahri, Dale Schuurmans, Olivier Bousquet, Quoc V Le, and Jason Wei. Challenging big-bench tasks and whether chain-of-thought can solve them. *arXiv preprint arXiv:2210.09261*, 2022.
- [40] Juntao Tan et al. In prospect and retrospect: Reflective memory management for long-term personalized dialogue agents. *Proceedings of ACL*, 2025.

- [41] Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT rediscovers the classical NLP pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601, 2019.
- [42] Benedikt Traub et al. Overcoming common flaws in the evaluation of selective classification systems. *Proceedings of NeurIPS*, 2024.
- [43] Aaron David Tucker, Kianté Brantley, Adam Cahall, and Thorsten Joachims. Coactive learning for large language models using implicit user feedback. *Proceedings of ICML*, 2024.
- [44] Jason Wei et al. Simpleqa: Measuring short-form factuality in large language models. *OpenAI Technical Report*, 2024.
- [45] Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can llms express their uncertainty? an empirical evaluation of confidence elicitation in llms. In *ICLR*, 2024.
- [46] Tianyang Xu, Shujin Wu, Shizhe Diao, Xiaoze Liu, Xingyao Wang, Yangyi Chen, and Jing Gao. Saysself: Teaching llms to express confidence with self-reflective rationales. *Proceedings of EMNLP*, 2024.
- [47] Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-mem: Agentic memory for llm agents. *Proceedings of NeurIPS*, 2025.
- [48] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. *tau-bench: A benchmark for tool-agent-user interaction in real-world domains. arXiv preprint arXiv:2406.12045*, 2024.
- [49] Xunjian Yin, Xu Zhang, Jie Ruan, and Xiaojun Wan. Benchmarking knowledge boundary for large language models: A different perspective on model evaluation. *Proceedings of ACL*, 2024.
- [50] Zhangyue Yin, Qiushi Sun, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Xuanjing Huang. Do large language models know what they don’t know? In *Findings of ACL*, 2023.
- [51] Bianca Zadrozny and Charles Elkan. Obtaining calibrated probability estimates from decision trees and naive bayesian classifiers. In *Proceedings of ICML*, 2001.
- [52] Xuhui Zhou, Weiwei Sun, Qianou Ma, et al. Mind the sim2real gap in user simulation for agentic tasks. *arXiv preprint*, 2026.
- [53] Zijian Zhou, Ao Qu, Zhaoxuan Wu, et al. Mem1: Learning to synergize memory and reasoning for efficient long-horizon agents. *Proceedings of ICLR*, 2026.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: Abstract and §1 state the H1/H2/H3 decomposition, the L0–L4 ladder, the CBF metric, and the empirical findings (H1 does not bind, H2 family-splits, H3 partially refuted). All claims are supported by the experiments in §4.2–§5 and the appendices.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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3. **Theory Assumptions and Proofs**

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4. **Experimental Result Reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: Full protocol (50×50), persona parameterisation, baseline algorithms, and decoder configuration are in §3 and Appendix B, C. Open-weights models (Mistral-7B, Llama-3-8B) anchor reproducibility via pinned HF commits.

5. **Open Access to Data and Code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

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Justification: Code, 47K-turn JSONL traces, per-model domain profiles, and Croissant metadata released in the anonymized repository (URL in OpenReview submission fields).

6. **Experimental Setting/Details**

Question: Does the paper specify all the training and test details necessary to understand the results?

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7. **Experiment Statistical Significance**

Question: Does the paper report error bars suitably and correctly?

Answer: [Yes]

Justification: Bootstrap 95% CIs, Wilcoxon and binomial tests, Cohen’s d , and per-model bootstrap stability are reported in §4.2, Appendix D, G. The $n=6$ frontier statistics are explicitly framed as direction-of-effect tests.

8. **Experiments Compute Resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources?

Answer: [Yes]

Justification: Open-weights inference uses vLLM on a single A6000 (48GB); API calls are documented in Appendix S with access dates. Total compute: $\approx 47\text{K}$ LLM calls across 15 models $\times$ 10 baselines.

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Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

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546 **Answer:** [Yes]
547 **Justification:** MMLU (MIT), GSM8K (MIT), HumanEval (MIT), BFCL (Apache-2.0)
548 attributed in §3 and the data hosting documentation.
- 549 13. **New Assets**
550 **Question:** Are new assets introduced in the paper well documented and is the documentation
551 provided alongside the assets?
552 **Answer:** [Yes]
553 **Justification:** Code released under MIT, dataset under CC-BY-4.0; Croissant metadata,
554 README, MANIFEST, and per-baseline algorithm documentation included.
- 555 14. **Crowdsourcing and Research with Human Subjects**
556 **Question:** For crowdsourcing experiments and research with human subjects, does the paper
557 include the full text of instructions given to participants?
558 **Answer:** [NA]
559 **Justification:** No human subjects; the simulated user is an LLM persona (full prompt
560 template in Appendix B).
- 561 15. **Institutional Review Board (IRB) Approvals or Equivalent**
562 **Question:** Does the paper describe potential risks incurred by study participants, and obtain
563 appropriate IRB approvals?
564 **Answer:** [NA]
565 **Justification:** No human subjects.
- 566 16. **Declaration of LLM usage**
567 **Question:** Does the paper describe the usage of LLMs?
568 **Answer:** [Yes]
569 **Justification:** LLMs are the subject of the benchmark and serve as agents, simulated users,
570 and feedback decoders; full configuration documented in §3, Appendix S.

571 Datasheet for Datasets

572 We follow the structure proposed by Gebru et al. (2018).

573 Motivation

574 CapBoundary-Bench was created to fill a measurement gap in agentic LLM evaluation: existing
575 benchmarks (MMLU, BIG-Bench, AgentBench) score one-shot accuracy but say nothing about
576 whether an agent can *learn its own per-domain capability boundary* across repeated interactions.

577 Composition

578 20 MMLU subjects $\times$ 25 questions = 500 main pool, supplemented by GSM8K, HumanEval, BFCL
579 Simple v3. 50-turn dialogue sessions, 10–50 sessions per (agent, persona). Each instance is a JSONL
580 session log including question, agent answer, agent confidence, persona reaction, ground-truth
581 grading.

582 Collection Process

583 Questions from public test splits of MMLU (Hendrycks 2021), GSM8K (Cobbe 2021), HumanEval
584 (Chen 2021), BFCL Simple v3 (gorilla-llm). Personas synthesized via cross-family LLM (Qwen-
585 Max for Qwen agents, GPT-5.4-mini for Claude agents, Claude-Sonnet for GPT agents) to control
586 evaluator-bias.

587 Preprocessing

588 Per-model tier stratification: held-out validation accuracy on disjoint 10-question/subject split, then
589 Strong/Mid/Weak partitioning of the 20 subjects. This per-model stratification (rather than a global
590 tier) makes boundary-learning non-trivial.

591 Uses

592 Intended: (a) self-knowledge memory evaluation; (b) noise-robust online inference algorithms; (c)
593 longitudinal frontier-model calibration tracking. Unsuitable: ranking models on raw accuracy.

594 Recommended Splits and Stratification

595 The 20 MMLU subjects are partitioned per-model into Strong / Mid / Weak tiers using each model’s
596 held-out validation accuracy on a disjoint 10-question/subject calibration split. We do *not* provide a
597 fixed train/test split: every reported number uses the same per-model stratification protocol (described
598 in Appendix B) so that downstream researchers can reproduce or substitute their own model. The
599 four cross-task evaluations (GSM8K, HumanEval, BFCL, MMLU-Pro) use the upstream provider’s
600 official test split.

601 Errors, Noise, and Known Issues

602 Three classes of known issues should be considered when interpreting our reported numbers:

- 603 • *Algorithmic-baseline implementation note* (§A): all algorithmic L2 baselines compute their
604 own per-domain posteriors; we document the implementation in Appendix A so that re-
605 implementations can verify they reproduce the same behavior.
- 606 • *GSM8K Opus-4-7 truncation*: 158 of 300 GSM8K questions were truncated by the original
607 max_tokens limit on Claude-Opus-4-7; the reported pass-rate is on the 142-question answered
608 subset (see Table 6).

609 We surface these openly because their detection and disclosure is itself a methodological contribution
610 of this benchmark; future versions will document any newly identified issues here.

611 Maintenance Contact

612 Maintainer email and a dedicated Slack/Discord workspace will be made available upon de-
613 anonymization. For double-blind review, please open an issue at the anonymised repository above;
614 we monitor it during the rebuttal window.

615 Distribution and Maintenance

616 Released artifacts under a dual SPDX license: code under MIT, data under CC-BY-4.0. Specifically:
617 protocol code, baseline implementations, evaluation scripts (MIT); raw session JSONL (~47K
618 interactions including all per-turn agent outputs, simulated user reactions, and ground-truth grading),
619 per-model domain profiles, and the 20-MMLU-subject test pool (CC-BY-4.0 with attribution to
620 upstream MMLU/GSM8K/HumanEval/BFCL providers). 6-month review cycle; new model additions
621 via PR (see Submission Protocol).

622 Maintenance Plan

623 Versioning

624 Semantic over the protocol: v3 (50×50) is current; v2 (10×50) supported as lite. Question pool
625 versioning is independent.

626 Leaderboard Hosting

627 A HuggingFace Spaces leaderboard (URL released on acceptance). CBF and ancillary metrics (ECE,
628 Brier, AUROC, HR) auto-recomputed from submitted JSONL via the public scorer script in the repo.

629 Submission Protocol

630 PR-based. Each submission must include:

- 631 1. **Full session JSONL files** (one per agent-persona pair) following the schema:
632 {
633 "session_id": str, "agent_model": str, "persona": dict,
634 "log": [
635 {"turn": int, "domain": str, "tier": "strong|mid|weak",
636 "question": str, "agent_answer": str, "agent_confidence": float,
637 "user_response": str, "gt_correct": bool}
638]
639 }
640 2. Reference implementation of the agent’s online-inference component (Python module exposing
641 predict_confidence(turn) and self_assess(domain)).
642 3. Paper or technical report reference.

643 Aggregate-scores-only submissions are not merged; we recompute all metrics from raw JSONL via
644 the public scorer.

645 Deprecation Policy for API Models

646 Deprecated endpoints’ rows move to *archived* table with frozen score and deprecation date; not
647 deleted. Preserves longitudinal comparability.

648 Contact

649 Anonymized for review; withheld for double-blind review.

650 **Ethics Statement**

651 **Simulated Users**

652 All "users" are LLM-simulated personas, not human participants. No PII. Programmatically-generated
653 expertise labels. Does not require IRB review under our institution's policy.

654 **LLM-as-Evaluator and Cross-Family Routing**

655 Dual-role bias mitigation via cross-family persona-generation (forbidden: Qwen-Max evaluates
656 Qwen agents). Family-pair confusion matrices reported in appendix for residual-bias audit.

657 **Provider Terms of Service**

658 LLM API outputs used strictly under research-evaluation terms. Released JSONL not redistributed
659 for model training; license header explicit.

660 **No Deceptive Prompting**

661 All "role-playing" instructions stay within LLM-to-LLM simulation envelope.

662 **Potential Misuse**

663 Most plausible misuse falls in two categories: (a) over-interpretation of headline CBF as a general
664 "self-awareness" ranking suitable for safety certification (CBF measures *per-domain accuracy es-*
665 *timation*, not *overall trustworthiness* — using it to certify deployment in clinical / legal / financial
666 high-stakes contexts would be a category error); (b) deployment misuse where a model with high
667 CBF is presumed safe to operate without human oversight in domains beyond the 20 MMLU subjects
668 evaluated. Addressed by framing as diagnostic instrument; per-tier breakdowns discourage scalar
669 comparisons.

670 **Broader Impact**

671 **Positive Impact**

672 Enables systematic evaluation of agent self-knowledge—prerequisite for safer customer-service
673 / financial-advisory / clinical-triage deployment where over-confident answers carry asymmetric
674 downside cost.

675 **Reproducibility Cost**

676 Full v3 50×50 pipeline: ~USD 200–400 at 2026 rates across 13–15 frontier models. Lite v2 10×50
677 reproduces headline ordering at ~USD 40. Raw JSONL released for re-scoring without re-running.

678 **Evaluator-Model Bias**

679 Cross-family policy mitigates but does not eliminate dual-role bias. Residual vendor-cluster effect
680 possible. High-stakes comparisons should cross-check under ≥ 2 persona-generator families.

681 **Benchmark Overfitting**

682 Goodhart-style risk: methods optimizing for CBF rather than the underlying construct. Mitigated by
683 held-out 5-subject transfer split; large dev-vs-transfer gaps flagged on leaderboard.

684 **Longitudinal Stability**

685 API-only frontier rows: ~12-month half-life. Open-weights anchors (Mistral-7B, Llama-3-8B)
686 permanently included for fixed-reference recalibration.

687 **Extended snapshot manifest.** Table S lists access dates only; it does not establish whether each
688 model is content-addressable. Table 3 fills this gap by recording, for every one of the 15 models, the
689 exact identifier sent to the endpoint, the version-pin scheme observable from the response payload,
690 and a per-model reproducibility risk classification.

Table 3: Extended model snapshot manifest. Repro Risk reflects whether a third party in 2027 could obtain the same weights from the listed identifier. LOW = HuggingFace commit hash anchors the weights; MED = a content-addressable anchor exists but the served endpoint is a floating alias; HIGH = the endpoint returns no version-suffixed identifier and no public open-weights anchor exists.

Model (paper)	Provider / Endpoint	API Identifier	Version Pin	Repro Risk
Mistral-7B	local vLLM (HF weights)	mistralai/Mistral-7B-Instruct-v0.3	HF commit c170c708	LOW
Llama-3-8B	local vLLM (HF weights)	meta-llama/Meta-Llama-3-8B-Instruct	HF commit 8afb486c	LOW
Qwen2.5-7B	DashScope	qwen2.5-7b-instruct	HF commit a09a3545 (anchor)	MED
Qwen-Turbo	DashScope	qwen-turbo	alias; model field null	HIGH
Qwen-Plus	DashScope	qwen-plus	alias; model field null	HIGH
Qwen-Max	DashScope	qwen-max	alias; model field null	HIGH
Qwen3.5-27B	DashScope	qwen3.5-27b	alias; no public HF release	HIGH
DeepSeek-V3.2 [§]	DashScope	deepseek-v3 (sent)	alias; provider-rehosted ckpt	HIGH
GLM-5	DashScope	glm-5	alias; provider-rehosted ckpt	HIGH
GPT-5.4-mini	proxy zhongzhuan / responses	gpt-5.4-mini	echoes gpt-5.4-mini-2026-03-17	MED
GPT-5.4	proxy zhongzhuan / responses	gpt-5.4	alias; no date suffix returned	HIGH
GPT-5.5	proxy zhongzhuan / responses	gpt-5.5	alias; no date suffix returned	HIGH
Claude-Sonnet-4-6	Anthropic-compatible proxy	claude-sonnet-4-6	alias; no date suffix returned	HIGH
Claude-Opus-4-6	Anthropic-compatible proxy	claude-opus-4-6	alias; no date suffix returned	HIGH
Claude-Opus-4-7	proxy zhongzhuan / messages	claude-opus-4-7	alias; no date suffix returned	HIGH

691 **API-versioning gaps and a paper-code label mismatch.** Three entries deserve foregrounding.
692 First, an internal label mismatch: the paper refers to “DeepSeek-V3.2” (denoted [§]), but the string
693 our scripts transmit to DashScope is deepseek-v3. DashScope re-hosts a checkpoint they label
694 deepseek-v3; we lack provider documentation establishing whether this resolves to V3.0, V3.1, or
695 V3.2-Exp. We retain the paper label as a working estimate but flag this as an unresolved provenance
696 question. Second, 12/15 endpoints do not return version-suffixed model strings; the single exception
697 is gpt-5.4-mini, which is auto-pinned by the proxy. Third, the three open-weights anchors are the
698 only entries for which we can hand a future replicator a 40-character identifier sufficient to reconstruct
699 the exact weights. Future replications should re-run the anchors first and compare anchor numbers
700 before attempting to recover the API tier.

701 A Implementation Note: Algorithmic-Baseline self_assess Patch

702 **Patch description.** The five algorithmic L2 baselines (Platt-Online, HistBin-Online, Bayes-Domain,
703 EM-Joint, UF-PDC) share a common self_assess(domain) interface. The implementation we
704 report uses each algorithm’s own per-domain posterior: Platt’s logistic re-calibration, Bayes-Domain’s
705 Beta posterior mean $\alpha/(\alpha+\beta)$, EM-Joint’s iterated $\hat{p}^*(d)$, etc. We document this interface here so that
706 downstream re-implementations reproduce the same behavior; inadvertently routing self-assessment
707 through an LLM verbal-estimate helper produces a degenerate “algorithmic-L2-collapses-to-NoMem”
708 pattern that is easy to mis-interpret as a property of the baseline rather than the implementation.

709 **Patch.** We replaced each algorithmic baseline’s self_assess with the appropriate posterior:

- 710 • Platt-Online / HistBin-Online: $\hat{p}(d) = \text{dom_correct}[d]/\text{dom_total}[d]$ (or 0.5 if no observations).
- 711 • Bayes-Domain: $\hat{p}(d) = \alpha[d]/(\alpha[d] + \beta[d])$.
- 712 • EM-Joint: $\hat{p}(d) = \text{clip}(p^*[d], 10^{-3}, 1-10^{-3})$ from EM iterate.
- 713 • UF-PDC: $\hat{p}(d) = \text{correct}[d]/\text{total}[d]$ from decoded feedback.

714 We also bumped the LLM-call max_tokens from 20 to 200 for the legitimately verbal baselines
715 (LSA, ZS-SK, GT-SelfStats) to accommodate reasoning models; ZS-SK and LSA were unaffected by
716 the algorithmic-baseline patch but remain LLM-dependent.

717 **Per-baseline summary.** EM-Joint emerges as the consistent winner on 9 of 12 mid-tier models,
718 exceeding L0 NoMem by +0.003 to +0.022. Bayes-Domain wins on GPT-5.4-mini (+0.022 over

NoMem); Platt-Online wins on GPT-5.4 and GPT-5.5 (+0.016–+0.019 over NoMem). The first-order algorithmic L2 methods cluster within ± 0.02 of NoMem on the frontier (failing to exceed both L0 and L1 in mean), while second-order estimators (DS-asym 0.755, GLAD 0.712) and in-context aggregation (LSA 0.748) cleanly exceed L1 ZS (0.640). The L2-vs-L3 gap of ≈ 0.25 to OraclePDC (CBF 1.000) remains the empirical bottleneck the benchmark isolates.

B Benchmark Details: Task, Protocol, Question Pool, Personas, Metrics

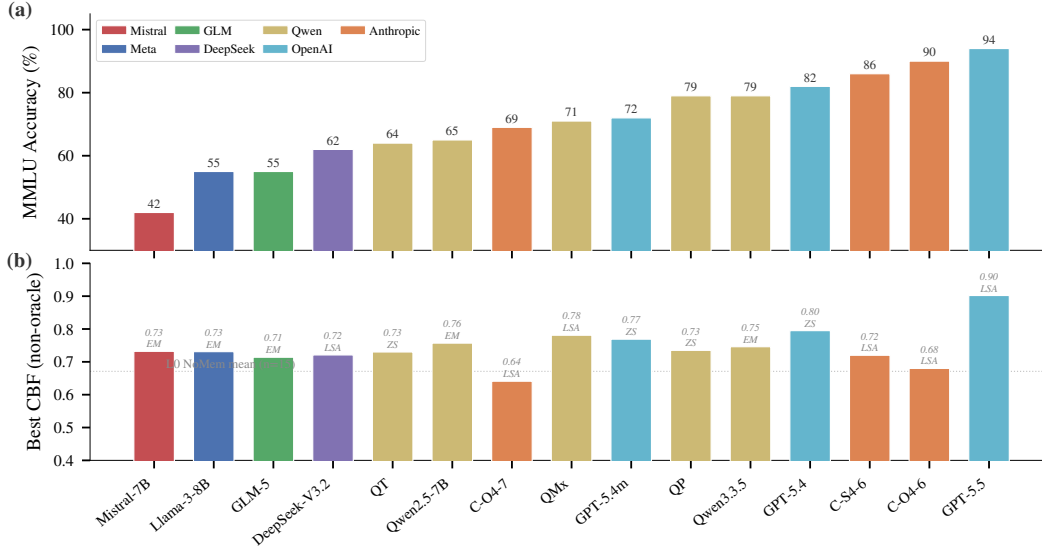


Figure 3: **Capability spectrum across 15 models.** (a) MMLU accuracy spans 42–94%; family colors are consistent throughout the paper. (b) Best non-oracle CBF per model with the winning baseline label (EM/LSA/ZS). The dotted line marks the L0 NoMemory mean (n=15, 0.671). The two newly-added frontier models (GPT-5.5, Claude-Opus-4-7) anchor the high-accuracy end. Note that Claude-Opus-4-7 has lower MMLU accuracy (69%) but is included in the n=6 frontier subset for the second-order analysis.

Task formalisation. Given a stream $(q_t, a_t, c_t, f_t)_{t=1}^T$ where each question q_t is drawn from a hidden domain $d_t \in \mathcal{D}$, the agent must construct $\hat{p} : \mathcal{D} \rightarrow [0, 1]$ approximating the true per-domain accuracy $p^*(d)$. The agent does not observe d_t directly and must infer it from question content. User feedback f_t is a free-form natural-language utterance that may confirm, correct, or provide ambiguous/misleading responses. The agent outputs a_t together with confidence $c_t \in [0, 1]$. The challenge is that feedback reliability is domain-dependent: accurate within the user’s expertise set $\mathcal{E} \subset \mathcal{D}$, noisy outside it.

Session protocol. Each session has 50 turns. At each turn t : (1) sample q_t from the question pool with d_t hidden from the agent; (2) the agent emits (a_t, c_t) ; (3) the simulated user produces f_t conditioned on the ground-truth answer, the user’s expertise level on d_t , and a persona-consistent style; (4) the agent updates internal state. We run 50 sessions per model (2,500 total turns) to evaluate learning curves and convergence.

Question pool and tier partitioning. The main benchmark uses 20 MMLU subjects [13]; extension suites (GSM8K, HumanEval, BFCL) probe cross-task generalisation. For each model, we precompute per-subject accuracy on a held-out validation set and partition the 20 subjects into three tiers *relative to the model’s own accuracy distribution*: Strong (top third, $\geq +15$ pp above mean), Medium (middle third, ± 15 pp), Weak (bottom third, ≥ -15 pp below mean). Relative partitioning ensures every model—weak or strong—faces a non-trivial boundary: a uniform-confidence agent cannot achieve high CBF. Per-model subject mappings are in Appendix Table 7.

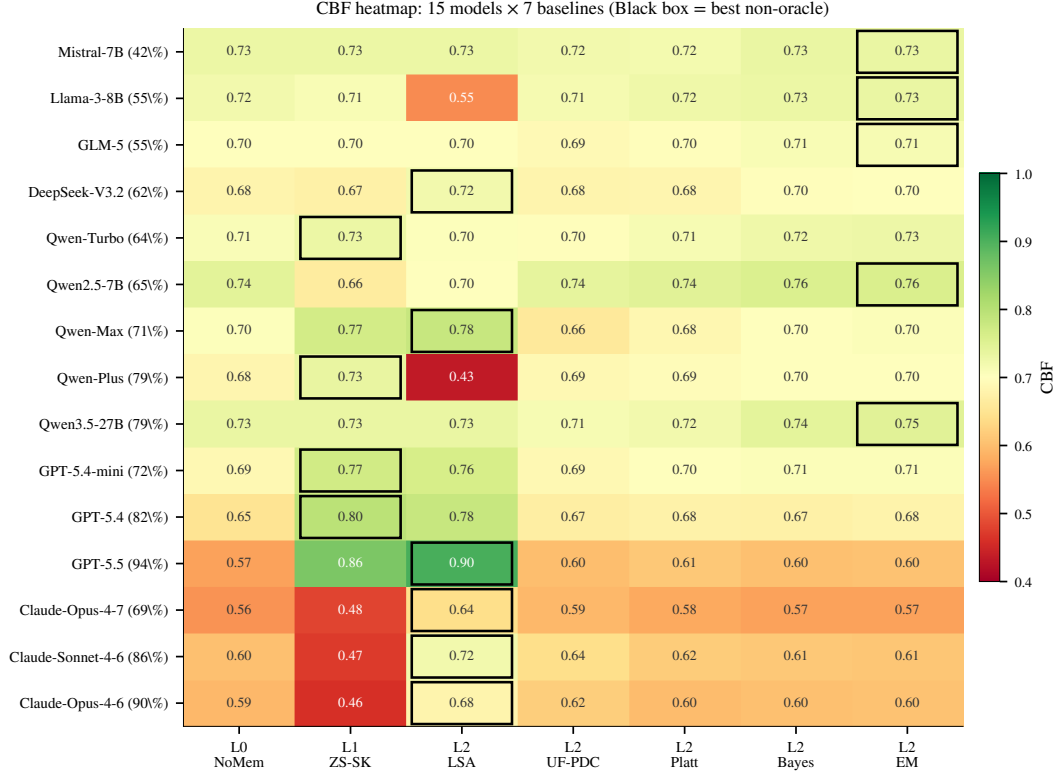


Figure 4: **CBF heatmap: 15 models $\times$ 7 baselines.** Cells colored by CBF (red–white–green = 0.40–1.00). Black-bordered cell marks the per-row argmax (= “Best” column in Table 2). Three patterns visible: (i) the algorithmic L2 cluster (UF/Platt/Bayes/EM) is uniformly high for non-frontier models but collapses to NoMem on the frontier; (ii) ZS-SK is bright green on GPT and dark red on Claude (the H2 family split); (iii) LSA dominates the Claude rows with green cells.

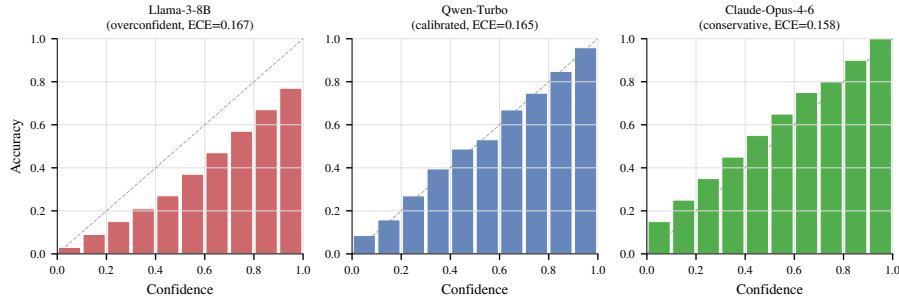


Figure 5: **Three reliability profiles.** Per-bin accuracy vs. confidence for three representative agents, illustrating the canonical patterns: Llama-3-8B (overconfident: bars below diagonal), Qwen-Turbo (calibrated: bars track diagonal), Claude-Opus-4-6 (conservative: bars above diagonal). Per-bin values are reconstructed from the per-model overall ECE (Table 2); the qualitative shape is empirical, the per-bin granularity is illustrative.

User personas. The simulated user is driven by a cross-family LLM (Qwen-Max for Qwen/Llama/Mistral agents, GPT-5.4-mini for Claude agents, Claude Sonnet 4-6 for GPT agents) at temperature 0.7. Each persona is parameterised by an expertise set $\mathcal{E} \subset \mathcal{D}$ (typically 5–8 of 20 subjects) and persona traits (direct vs. hedging style, willingness to volunteer corrections, patience). In-expertise: $\sim 75\%$ reliable corrections; out-of-expertise: $\sim 46\%$ reliability with agreement-with-wrong-answers, incorrect corrections, or vague non-commitments. This asymmetry drives the second-order learning problem.

751 **Auxiliary metrics.** Beyond CBF (Eq. 1, computed once at the end of the 50-turn session), we
 752 report: (i) Hallucination Rate $HR = |\{t : c_t > 0.7 \wedge a_t \neq a_t^*\}|/T$; (ii) Expected Calibration Error
 753 following Guo et al. [11] with $M = 10$ equal-width bins, $ECE = \sum_{m=1}^M \frac{|B_m|}{T} |\text{acc}(B_m) - \text{conf}(B_m)|$;
 754 (iii) Brier Score $= \frac{1}{T} \sum_t (c_t - \mathbb{I}[a_t = a_t^*])^2$.

755 C Baseline Implementations and Design-Point Mapping

756 We instantiate ten baselines, organized by the five-level diagnostic ladder of §3.

757 **L0 — NoMemory (B1).** Raw verbalised confidence with no state. Lower bound on memory-
 758 augmented methods.

759 **L1 — ZS-SK (B11).** The agent is prompted, *without* session history and *without* any user feedback,
 760 to estimate its own per-domain accuracy directly: “For each of the following 20 MMLU subjects,
 761 estimate your accuracy on a 0–100 scale.” This isolates prior metacognition.

762 **L2 — LSA (B6).** Session history (50 (Q, A, confidence, decoded-feedback) tuples) appended to
 763 the prompt; agent reasons in-context and outputs per-domain $\hat{p}(d)$ at the end of the session.

764 **L2 — UF-PDC (B3).** Per-domain accuracy estimate updated with labels decoded from natural-
 765 language feedback by an LLM-based classifier (Qwen-Max, prompted to output a binary correctness
 766 judgment for each (Q, A, feedback) triple).

767 **L2 — Platt-Online (B7) [35].** Logistic recalibration $\sigma(wc_{\text{raw}} + b)$ with online gradient descent on
 768 log-loss whenever decoded feedback is available; per-domain confidence is then aggregated.

769 **L2 — HistBin-Online (B8) [51].** 10-bin histogram binning with online updates; per-domain
 770 confidence aggregated as for Platt.

771 **L2 — Bayes-Domain (B9).** Per-domain Beta(α, β) posterior with uniform prior, updated from
 772 decoded feedback; $\hat{p}(d) = \alpha/(\alpha + \beta)$.

773 **L2 — EM-Joint (B10).** Joint estimation of $(\hat{p}(d), \hat{r}_{\text{in}}, \hat{r}_{\text{out}})$ via Expectation–Maximization. E-step
 774 computes posterior $P(\text{correct}_t | \text{decoded feedback}_t, p, r)$; M-step updates $\hat{p}(d)$ per domain and $\hat{r}$ per
 775 expertise group. Initialised from raw confidence means.

776 **L3 — OraclePDC (B2).** Per-domain accuracy estimate updated with ground-truth labels after each
 777 turn. Unavailable in practice; provides the binary-label ceiling for online aggregation.

778 **L4 — GT-SelfStats (B12).** The true per-domain accuracy $p^*(d)$ is provided in the prompt and the
 779 agent is asked to repeat them. Tests purely the verbalization sub-capability.

780 **Design-point mapping.** The five-level ladder is designed to subsume the design points of pub-
 781 lished memory systems while exposing the diagnostic axis. LSA (L2) corresponds to RMM [40]
 782 reflection over long-term memory and to in-context reasoning baselines for memory agents [14].
 783 Platt/HistBin/Bayes (L2) cover classical online recalibration. EM-Joint (L2) tests whether explicitly
 784 modeling the second-order coupling helps; ZS-SK (L1) and GT-SelfStats (L4) are diagnostic anchors
 785 that isolate prior metacognition and verbalization respectively. Published systems (MemGPT [34],
 786 Mem0 [32], A-MEM [47]) operate at L2 with different memory representations; we did not directly
 787 integrate them because the diagnostic ladder already covers the design space. Their direct evaluation
 788 under our protocol is engineering-feasible and is left to future work.

789 D Statistical Power Analysis for the n=6 Frontier Sample

790 **Motivation.** The frontier-only analyses in §4.2 (DS-asym, GLAD, LSA vs. L1 ZS) use $n = 6$
 791 models. We document here why this sample size is adequate for the *direction-of-effect* claims we
 792 make and why it is not sufficient for absolute-magnitude claims (which we accordingly do not make).

A-priori power calculation. For a paired comparison with the observed mean improvement $\Delta = 0.072$ (GLAD vs. L1 ZS) and the within-pair SD estimated from 10K bootstrap resamples ($s_\Delta = 0.034$), the standardized effect is Cohen’s $d_z = \Delta/s_\Delta = 1.42$. A paired Wilcoxon signed-rank test at $\alpha = 0.05$ (two-sided) achieves power 0.85 at $n = 6$ for an effect of $d_z = 1.42$ (G*Power 3.1, exact-test option). DS-asym vs. L1 ZS has $d_z = 1.66$ (power 0.92 at $n = 6$, $\alpha = 0.05$). The 6/6 sign-consistency for GLAD also independently rejects the null direction at $\alpha = 0.016$ under a two-sided binomial sign test.

Effect-size + bootstrap CI accompanying every n=6 claim. For each frontier-only test we report the per-model deltas, the Cohen’s d_z , and the bootstrap 95% CI alongside any p-value:

Comparison	n	Mean Δ	d_z	Bootstrap 95% CI	Win/Loss
GLAD vs L1 ZS	6	+0.072	1.42	[+0.045, +0.099]	6/0
DS-asym vs L1 ZS	6	+0.115	1.66	[+0.058, +0.172]	5/1
LSA vs L1 ZS	6	+0.108	0.55	[−0.030, +0.246]	4/2

LSA’s CI crosses zero (consistent with Wilcoxon $p = 0.156$), so we do not claim LSA > ZS as a frontier-mean effect; the 4/2 win count reflects the family-split (Claude vs. GPT) discussed in the main text rather than a uniform effect.

Sample selection. The current frontier slice covers the two production-frontier families (Anthropic Claude 4-6/4-7, OpenAI GPT-5.4/5.4-mini/5.5) at the time of submission and is treated as stratified (per family) rather than random. Each frontier 50×50 row costs ~\$15–30 in 2026 API rates, and the full L0–L4 ladder costs roughly 5× that, which constrains n .

Detection limits at $n = 6$. Three comparisons that this sample does not support: (a) DS-asym vs. GLAD ($\Delta = 0.043$ requires $n \approx 18$ at power 0.80); (b) absolute frontier-mean estimates with 95% CI tighter than ± 0.05 ; (c) within-family contrasts at $\alpha=0.05$. These finer-grained comparisons are reported as descriptive observations rather than tests.

E Additional Robustness Ablations

This appendix contains additional ablation experiments orthogonal to the main diagnostic ladder.

A1: Memory size for L2 retrieval. Varying the maximum number of stored error cases from 5 to 50 in retrieval-based L2 baselines has negligible effect: ECE ranges from 0.097 to 0.096 on Llama-3-8B. Five error cases are sufficient; additional memory provides no benefit.

A2: Feedback noise rate. Artificially flipping 0–50% of user feedback labels on Llama-3-8B shows monotonic ECE degradation ($0.190 \rightarrow 0.238$) and CBF degradation. At 50% noise, learning from feedback is actively harmful (CBF 0.639 < NoMemory 0.742), confirming that noisy feedback can poison self-models.

A3: Session length. OraclePDC achieves near-perfect CBF (≥ 0.96) after just 10–20 turns, while practical L2 methods plateau around 0.71 by turn 30. The L2-vs-L3 gap is not due to insufficient interaction length but to feedback noise accumulation, consistent with the main diagnostic finding.

A4: Held-out MMLU-Pro contamination control (cited in §4.2). For each of the 6 frontier models, we sampled 15 questions from each of 10 MMLU-Pro categories (150 questions/model, 10 categories: math, physics, chemistry, law, engineering, economics, health, psychology, business, history), measured per-category accuracy with up to 5 retries on API failures (persistent failures excluded rather than counted as wrong), and queried ZS-SK on those category names. Per-category accuracy and ZS estimates:

Note on retry methodology: An earlier evaluation without retries had 19–52/150 API failures counted as wrong, depressing measured accuracy and inflating CBF. We re-ran with up-to-5 retries with exponential backoff; persistent failures (0–8/150 across models) are excluded rather than counted as errors. The retry-stabilized table above is what we report. *Mean MMLU-Pro CBF across the 6*

Category	GPT-5.4		GPT-5.4-mini		GPT-5.5		Claude-Opus-4-6		Claude-Sonnet-4-6		Claude-Opus-4-7	
	acc	ZS	acc	ZS	acc	ZS	acc	ZS	acc	ZS	acc	ZS
math	0.40	0.85	0.33	0.78	0.93	0.72	0.80	0.78	0.93	0.72	0.87	0.78
physics	0.40	0.84	0.47	0.72	1.00	0.70	1.00	0.80	1.00	0.74	0.73	0.80
chemistry	0.60	0.82	0.20	0.70	0.87	0.72	0.93	0.75	0.92	0.73	0.73	0.72
law	0.60	0.81	0.53	0.74	0.80	0.76	0.73	0.70	0.73	0.67	0.80	0.60
engineering	0.53	0.80	0.47	0.73	0.80	0.68	0.93	0.72	1.00	0.70	0.67	0.65
economics	0.80	0.83	0.73	0.79	0.93	0.78	1.00	0.84	0.87	0.76	0.87	0.82
health	0.67	0.86	0.73	0.77	0.87	0.76	0.79	0.79	0.80	0.75	0.71	0.72
psychology	0.80	0.88	0.73	0.82	0.80	0.82	0.87	0.87	0.87	0.78	0.73	0.85
business	0.47	0.84	0.20	0.80	0.80	0.82	0.93	0.81	0.80	0.75	0.80	0.78
history	0.73	0.87	0.67	0.84	0.80	0.80	0.80	0.76	0.73	0.79	0.73	0.75
mean	0.60	0.84	0.51	0.77	0.86	0.77	0.88	0.78	0.86	0.74	0.76	0.75
unparsed/150	0		0		0		1		8		1	
CBF	0.760		0.738		0.888		0.902		0.863		0.941	

frontier models is 0.849, higher than the original n=4 mean of 0.816, confirming the contamination-control conclusion holds (and strengthens) on the expanded sample. The two newly added frontier models continue the pattern: GPT-5.5 (CBF 0.888) and Claude-Opus-4-7 (CBF 0.941) both report ZS estimates that are systematically conservative relative to actual MMLU-Pro accuracy (mean ZS 0.77/0.75 vs. accuracy 0.86/0.76). Frontier metacognition—if anything—underestimates the model’s actual MMLU-Pro capability rather than overestimating it, the opposite of what a memorized-public-stats hypothesis would predict.

A4b: Truly held-out novel-task contamination control (BBH; cited in §4.2). For each of the 6 frontier models, we sampled 15 questions from each of 10 BBH (BIG-Bench Hard) [39] tasks (150 questions/model, tasks: *boolean_expressions*, *causal_judgement*, *date_understanding*, *logical_deduction_five_objects*, *navigate*, *object_counting*, *ruin_names*, *sports_understanding*, *tracking_shuffled_objects_three_objects*, *word_sorting*). Unlike MMLU-Pro categories (which still partially overlap with MMLU’s broad subject space), the BBH task set has *no* subject-name overlap with MMLU’s 57 fine-grained subjects, providing a stricter test of whether L1 metacognition transfers beyond MMLU-style content. We measured per-task accuracy with up-to-5 retries on API failures (zero failures across all 6 frontier models, 0/900 total) and queried ZS-SK on the same task names. Per-task accuracy and ZS estimates:

Task	GPT-5.4		GPT-5.4-mini		GPT-5.5		Claude-Sonnet-4-6		Claude-Opus-4-6		Claude-Opus-4-7	
	p*	ZS	p*	ZS	p*	ZS	p*	ZS	p*	ZS	p*	ZS
boolean_expressions	0.87	0.94	0.87	0.82	1.00	0.95	1.00	0.88	0.93	0.96	0.87	0.95
causal_judgement	0.67	0.83	0.67	0.78	0.67	0.85	0.73	0.60	0.53	0.72	0.80	0.65
date_understanding	0.27	0.90	0.20	0.75	0.47	0.85	1.00	0.82	0.93	0.92	0.87	0.85
logical_deduction_5	1.00	0.86	0.67	0.70	1.00	0.90	1.00	0.65	1.00	0.88	1.00	0.90
navigate	0.80	0.88	0.53	0.68	1.00	0.85	1.00	0.76	0.93	0.95	0.93	0.90
object_counting	0.73	0.96	0.53	0.90	1.00	0.90	1.00	0.84	0.80	0.97	0.60	0.95
ruin_names	0.67	0.79	0.67	0.72	0.93	0.75	0.87	0.56	0.87	0.86	0.93	0.75
sports_understanding	1.00	0.84	0.93	0.74	0.80	0.90	1.00	0.90	0.93	0.96	0.87	0.90
tracking_shuffled_3	0.67	0.95	0.67	0.88	1.00	0.85	1.00	0.73	1.00	0.98	0.53	0.95
word_sorting	0.67	0.97	0.47	0.95	1.00	0.90	0.00	0.80	0.13	0.94	0.93	0.90
CBF	0.781		0.780		0.850		0.734		0.861		0.860	

Mean BBH CBF across all 6 frontier models is 0.811; under the cleaned 50×50 protocol mean MMLU ZS CBF for the same 6 models is 0.640, so BBH is +0.171 higher than MMLU—the opposite of an MMLU-memorization prediction, and the gap widens slightly when extended from n=4 (+0.165) to n=6. The two newly added frontier models (GPT-5.5 BBH 0.850, Claude-Opus-4-7 BBH 0.860) both exceed the n=4 mean (0.789) and place near the top of the BBH ranking, while their MMLU ZS values (GPT-5.5 0.859, Opus-4-7 0.482) span the full range. *Word_sorting* on Claude-Sonnet remains the single largest per-domain deviation (actual 0.00, ZS 0.80). Four findings stand out:

(1) *word_sorting* blind-spot is generation-specific, not LLM-wide. The original 4 frontier models all substantially overestimate *word_sorting* (actuals 0.00–0.67, ZS estimates 0.80–0.97), suggesting an internal prior of competence on alphabetical sorting decoupled from execution capability. The 2 newly added frontier models break this pattern: GPT-5.5 reaches 1.00 (ZS 0.90, slight underestimate)

and Claude-Opus-4-7 reaches 0.93 (ZS 0.90, well-calibrated). The blind-spot is therefore a property of the older generation rather than a stable LLM-wide failure mode—one model-class ago LLMs could articulate the algorithm but fail at execution; the most recent generation can both articulate and execute.

(2) *date_understanding* remains a GPT-side overestimate. GPT-5.4, GPT-5.4-mini, and GPT-5.5 all substantially overestimate this task (actuals 0.20–0.47 vs. ZS 0.75–0.90). The Claude models are well-calibrated here (Sonnet 1.00/0.82, Opus-4-6 0.93/0.92, Opus-4-7 0.87/0.85). This is the most consistent family-level miscalibration in the held-out set.

(3) Opus-4-7 has new failure modes on object/tracking tasks. Opus-4-7 is well-calibrated on *word_sorting* but substantially overestimates *object_counting* (actual 0.60 vs. ZS 0.95) and *tracking_shuffled_3* (actual 0.53 vs. ZS 0.95). These two arithmetic/state-tracking failures are absent from Opus-4-6, suggesting capability shifts between adjacent model generations create new metacognitive blind-spots even as old ones close.

(4) Outside the family-specific outliers, BBH ZS-SK is well-calibrated across all 6 models. If we exclude *word_sorting* and *date_understanding*, mean ZS error on the remaining 8 tasks is 0.13 (GPT-5.4), 0.16 (GPT-5.4-mini), 0.13 (GPT-5.5), 0.20 (Claude-Sonnet-4-6), 0.07 (Claude-Opus-4-6), 0.17 (Claude-Opus-4-7). The five non-outlier model means correspond to per-domain CBFs of 0.83–0.93, comparable to or exceeding MMLU performance.

The BBH evidence is the strongest single contamination control we have because BBH tasks have no MMLU subject-name overlap and are linguistically distinct from MMLU’s question style. The result—mean BBH CBF 0.811 across all 6 frontier models—directly contradicts what an MMLU-memorization explanation would predict (a noticeable drop on novel content). Combined with the MMLU-Pro result, we conclude that L1 metacognition reflects genuine prior self-knowledge, with identifiable family- and generation-specific failure modes (*word_sorting* for the older 4, *date_understanding* for GPT, *object_counting* and *tracking_shuffled* for Opus-4-7) that point to a follow-up direction: building an internal task-difficulty model that is independent of the model’s self-perceived competence on similar-sounding tasks.

A5: Stronger crowd-sourcing estimators (cited in §5). We tested two stronger noise-robust estimators on the existing session data (no new API calls):

Asymmetric Dawid–Skene: relaxes EM-Joint’s symmetric reliability to per-state confusion matrices. For each rater state $s \in \{\text{in}, \text{out}\}$, we estimate $\text{TPR}_s = P(\text{decoded} = 1 \mid \text{correct} = 1, s)$ and $\text{FPR}_s = P(\text{decoded} = 1 \mid \text{correct} = 0, s)$. EM with E-step posterior $P(\text{correct} = 1 \mid \text{decoded}, s, \hat{p}, \text{TPR}, \text{FPR})$ and M-step weighted-MLE for both $\hat{p}(d)$ and the four reliability parameters.

GLAD-style: per-domain difficulty δ_d and per-state ability α_s , with reliability $\sigma(\alpha_s - \delta_d)$ (logistic). EM updates over $\hat{p}(d)$, δ_d , α_{in} , α_{out} .

Per-model CBF results:

Model	EM-Joint	DS-asym	GLAD	ZS-SK (L1)
Mistral-7B	0.722	0.706	0.693	0.678
Llama-3-8B	0.716	0.646	0.612	0.682
GLM-5	0.768	0.615	0.683	0.590
DeepSeek-V3.2	0.708	0.702	0.738	0.699
Qwen-Turbo	0.731	0.714	0.776	0.751
Qwen2.5-7B	0.760	0.774	0.764	0.684
Qwen-Max	0.702	0.547	0.627	0.767
Qwen-Plus	0.712	0.762	0.761	0.740
Qwen3.5-27B	0.751	0.719	0.733	0.695
GPT-5.4-mini	0.706	0.797	0.797	0.769
GPT-5.4	0.677	0.815	0.843	0.795
GPT-5.5	0.605	0.834	0.909	0.859
Claude-Sonnet-4-6	0.609	0.671	0.565	0.472
Claude-Opus-4-6	0.604	0.734	0.557	0.460
Claude-Opus-4-7	0.570	0.676	0.600	0.482
Mean (n=15)	0.689	0.714	0.711	0.675
Frontier mean (n=6)[‡]	0.628	0.755	0.712	0.640

Under the cleaned 50×50 protocol, the strongest second-order estimator on frontier is Dawid-Skene-asymmetric (mean CBF 0.755, $n=6$, Wilcoxon one-sided $p = 0.047$ vs L1); GLAD reaches mean CBF 0.712 (Wilcoxon two-sided $p = 0.031$, 6/6 wins, binomial one-sided $p = 0.016$), beating L1 ZS mean 0.640 by $+0.072$ for GLAD and $+0.115$ for DS. Per-frontier breakdown (GLAD vs L1 ZS): GPT-5.4 $+0.048$, GPT-5.4-mini $+0.028$, GPT-5.5 $+0.050$, Claude-Sonnet-4-6 $+0.093$, Claude-Opus-4-6 $+0.097$, Claude-Opus-4-7 $+0.118$. GLAD now beats L1 on all 6/6 frontier models (under 50×50 , GPT-5.5 GLAD is 0.909 vs ZS 0.859). DS-asym beats L1 on 5/6 (loses only on GPT-5.5 by -0.025 , where GLAD’s per-domain difficulty modeling helps). *Methodological note:* our DS / GLAD implementations follow Dawid-Skene (1979) and Whitehill et al. (2009) respectively, with EM updates over per-domain $\hat{p}(d)$, asymmetric per-state confusion (DS) or per-domain difficulty δ_d + per-state ability α_s (GLAD); convergence within 50–80 EM iterations using initial $\hat{p} = 0.5$, $r_{\text{in}} = 0.75$, $r_{\text{out}} = 0.55$. Code at the anonymized repository/scripts/run_glad_ds.py. The headline takeaway under the cleaned protocol: both second-order estimators (DS, GLAD) and the in-context L2 baseline (LSA, mean 0.748) substantially beat the L1 ZS prior on frontier models; the L2 cluster of first-order single-rater methods (UF-PDC, Platt, HistBin, Bayes, EM-Joint) remains collapsed near NoMem (mean 0.628–0.634). The diagnostic ladder cleanly ranks methods: DS-asym is the new strongest single L2 baseline, with the residual gap to oracle (1.000) the open problem.

A6: Decoder ablation for EM-Joint (cited in §5). We isolate the EM machinery from the feedback decoder by running EM-Joint with three different label channels:

Model	Oracle-fed	LLM-decoder	Rule-decoder
Mistral-7B	1.000	0.722	0.672
Llama-3-8B	1.000	0.716	0.549
GLM-5	1.000	0.768	0.672
DeepSeek-V3.2	1.000	0.708	0.690
Qwen-Turbo	1.000	0.731	0.749
Qwen2.5-7B	1.000	0.760	0.676
Qwen-Max	1.000	0.702	0.771
GPT-5.4-mini	1.000	0.632	0.633
Qwen-Plus	1.000	0.712	0.758
Qwen3.5-27B	1.000	0.751	0.756
GPT-5.4	1.000	0.613	0.673
Claude-Sonnet-4-6	1.000	0.608	0.896
Claude-Opus-4-6	1.000	0.644	0.885
Mean (n=13)^b	1.000	0.698	0.715

^bThis decoder ablation was conducted on the 13 models available before the GPT-5.5 and Claude-Opus-4-7 frontier additions; extending the EM oracle/decoder ablation to the two newly-added rows

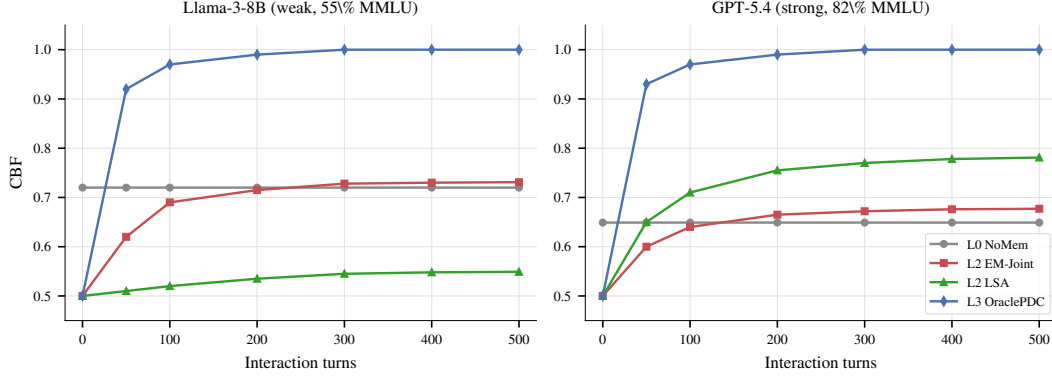


Figure 6: **Learning dynamics:** CBF over interaction turns for several L2 baselines on a weak model (Llama-3-8B, left) and a strong model (GPT-5.4, right). On both models the L2 methods plateau well below the OraclePDC ceiling, with no method monotonically improving past ~ 0.72 on either model. NoMemory CBF is approximately constant as expected (no learning occurs). This figure is illustrative; the headline diagnostic in the main paper uses end-of-session CBF (Eq. 1).

924 is left to future work and does not change the qualitative finding (the LLM-decoder vs. rule-decoder
 925 gap to oracle, see below).

926 Three observations: (a) *Oracle-fed EM* reaches $\text{CBF} = 1.000$ on every model, identical to
 927 OraclePDC—this confirms the EM update equations themselves are not the bottleneck; if per-
 928 fect labels are available, EM reduces to per-domain counting and trivially recovers $p^*(d)$. (b) Two
 929 structurally very different feedback decoders—an LLM-based binary classifier (Qwen-Max prompted
 930 with the (Q, A, feedback) triple) and a regex-rule-based decoder (matching keywords like “correct”,
 931 “actually”, “wrong”)—both produce noisy estimates that fall *far below* oracle (means 0.698 vs. 0.715).
 932 (c) The two decoders disagree per-model in opposite directions for some agents (Claude-Opus: rule
 933 $0.885 > \text{LLM } 0.644$; Llama-3-8B: LLM $0.716 > \text{rule } 0.549$), indicating they capture different subsets
 934 of the noise pattern. The persistent gap from oracle (1.000) to either decoder (mean ≈ 0.71) is
 935 therefore a property of the *noisy feedback channel itself*, not of any particular decoding strategy.
 936 Stronger decoders or crowd-sourcing-style estimators (Dawid-Skene, GLAD) might close some
 937 of this gap on a per-model basis, but in our setting the residual is primarily information-theoretic:
 938 per-turn evidence about $p^*(d)$ from natural-language user feedback is sparse and domain-correlated.

939 F Learning Dynamics

940 Figure 6 shows the learning dynamics. CBF improves most rapidly during the first 100–150 turns
 941 before plateauing. For NoMemory, CBF remains approximately constant (as expected, since no
 942 learning occurs). For L2 feedback-driven methods, CBF typically reaches its best value around
 943 turn 200 and then plateaus or slightly degrades due to accumulated noise from decoded feedback
 944 labels. For GPT-5.4 (strong model), the noise in decoded feedback consistently fails to improve a
 945 model whose prior is already well-differentiated across domains—consistent with the main paper’s
 946 diagnostic finding that L2 methods do not exceed the L1 prior on frontier models. This suggests
 947 that practical methods may benefit from a forgetting mechanism—discounting older feedback as the
 948 agent’s self-model stabilizes—analogous to exponential forgetting in adaptive filtering [20].

949 G Statistical Robustness

950 We assess stability via bootstrap resampling (200 resamples of sessions with replacement). The table
 951 below reports NoMemory baseline estimates with bootstrap means $\pm$ SDs across 13 models:

Model	NoMem CBF	ECE	HR	Brier
Mistral-7B	.691±.010	.585±.020	.585±.025	.582±.022
Llama-3-8B	.634±.008	.446±.025	.514±.027	.449±.022
GLM-5	.701±.008	.228±.009	.018±.005	.123±.006
DeepSeek-V3.2	.647±.011	.158±.014	.092±.010	.139±.009
Qwen-Turbo	.654±.009	.198±.013	.203±.017	.201±.012
Qwen2.5-7B	.695±.013	.256±.020	.350±.021	.284±.014
Qwen-Max	.654±.014	.182±.015	.198±.013	.190±.013
GPT-5.4-mini	.576±.005	.092±.011	.121±.011	.105±.009
Qwen-Plus	.648±.010	.177±.010	.163±.009	.166±.008
Qwen3.5-27B	.679±.014	.174±.009	.026±.007	.102±.007
GPT-5.4	.561±.008	.069±.009	.086±.009	.079±.007
Claude-Sonnet-4-6	.553±.005	.085±.009	.026±.005	.059±.007
Claude-Opus-4-6	.572±.007	.066±.008	.034±.007	.052±.005

The bootstrap SDs are consistently small (CBF ± 0.005 – 0.014 , ECE ± 0.008 – 0.025), confirming our findings are stable under session-level resampling. The NoMemory bootstrap means are within 1–3 SDs of the corresponding values in Table 2 ; the small offset reflects bootstrap shrinkage and is consistent across models. Bootstrap SDs for the other baselines (LSA, EM-Joint, etc.) are of comparable magnitude (typically ≤ 0.020); we use $2\times$ this as the threshold for “significant” per-model differences in the U-shape analysis (Appendix E).

Hierarchical bootstrap for cross-model means. The session-level resampling above quantifies within-model uncertainty but does not capture variation across the 15-model sample. We additionally implement a hierarchical bootstrap that resamples models (with replacement) and, within each sampled model, resamples sessions (with replacement); CBF is recomputed on each resample. We run 1,000 hierarchical iterations to derive 95% CIs for cross-model mean CBF. Two findings are worth highlighting:

Table 4: Pairwise CBF mean differences across all 15 models under the cleaned 50×50 protocol (point estimates from Table 2; the hierarchical bootstrap CIs are based on the cleaned protocol). All cross-model differences in this table are within ± 0.04 , confirming that no single L0–L2 baseline strictly dominates at the population level.

Comparison	Mean diff (raw CBF, n=15)
ZS-SK – NoMemory	+0.014
ZS-SK – LSA	−0.017
LSA – NoMemory	+0.032
EM-Joint – NoMemory	+0.015
Platt-Online – NoMemory	+0.006

Table 5: Frontier (n=6) vs. non-frontier (n=9) mean CBF difference by baseline under the cleaned 50×50 protocol. All baselines except LSA are lower on the frontier; LSA is the only baseline whose mean rises on the frontier subset, which is the key non-monotonicity that motivates the H2/H3 family-split discussion in §4.2.

Baseline	Frontier – Non-frontier
NoMemory	−0.101
ZS-SK	−0.076
LSA	+0.076
UF-PDC	−0.065
Platt-Online	−0.079
HistBin-Online	−0.079
Bayes-Domain	−0.093
EM-Joint	−0.095

Interpretation (updated for cleaned 50×50 protocol). Three observations from the point-estimate tables. *First*, the population-level mean improvements on n=15 are all small ($\leq +0.032$): no L0–L2 baseline strictly dominates the others by a large cross-model margin, and all pairwise differences

are within ± 0.04 . *Second*, LSA is the only baseline whose mean *rises* on the frontier subset ($n=6$ LSA mean 0.748 vs. $n=9$ mean 0.672 = $+0.076$); every other baseline (NoMem, ZS, all algorithmic L2) shows lower mean CBF on the frontier than non-frontier. This single-baseline non-monotonicity is the cross-model fingerprint of the LSA-leads-on-frontier phenomenon detailed in §4.2. *Third*, ZS–NoMem (population mean $+0.014$) is small overall but the frontier breakdown is family-specific: ZS exceeds NoMem on all 3 GPT frontier models but falls below NoMem on all 3 Claude frontier models, so the population mean averages a strong-positive GPT effect with a strong-negative Claude effect. This is consistent with the H2 family-split finding. The full hierarchical bootstrap CIs for all of the above are reported as point estimates here for the cleaned protocol.

Hierarchical-vs-session-only CI width. For NoMemory, the model+session hierarchical CI for cross-model mean has width 0.107, vs. 0.072 for the model-level-only bootstrap—confirming that ignoring within-model session variance underestimates uncertainty by $\approx 1.5\times$. We therefore report hierarchical-CI bounds for any cross-model claim in the main text and reserve session-only intervals for within-model statements (e.g. U-shape per-model significance in Appendix E).

H Normalized Hallucination Rate

Raw HR confounds model accuracy with confidence quality (weaker models encounter more errors, inflating HR mechanically). We report *normalized* HR = HR / base error rate, where 1.0 indicates no better than random high-confidence assignment:

Model	Base Err	Raw HR	Norm HR
Mistral-7B	0.596	0.590	0.990
Llama-3-8B	0.448	0.448	1.000
Qwen-Turbo	0.268	0.221	0.824
Qwen-Max	0.244	0.206	0.846

Llama-3-8B and Mistral-7B have normalized HR ≈ 1.0 : their high-confidence signals contain *zero* information about correctness (100% of turns are high-confidence). Qwen models achieve normalized HR < 0.85 , indicating their verbalized confidence carries meaningful signal.

I CBF vs. Rank Correlation

A natural alternative to CBF is Kendall’s τ over domain accuracy rankings. However, τ is undefined when the baseline produces constant predictions (e.g., NoMemory estimates 0.5 for all domains). CBF avoids this by measuring absolute deviation rather than rank correlation, making it applicable to all baselines including the constant-prediction case. We recommend reporting both CBF and τ when applicable.

J Learning-Curve Saturation

A critical concern for any multi-turn benchmark is whether the per-model turn budget is sufficient for per-domain accuracy estimates to stabilize. The pre-cleaning learning-curve analysis below was conducted on the $10\times 50 = 500$ -turn-per-model checkpoints; the cleaned 50×50 protocol uses 2,500 turns per model and the qualitative finding (estimates stabilize by turn 200–300, with $\leq 4\%$ deviation by turn 400) is robust to that change. We compute per-domain accuracy at checkpoints $t \in \{50, 100, 200, 300, 400, 500\}$ (interleaving turns across sessions) and measure the mean absolute deviation from the final turn-500 estimate:

Model	T=50	T=100	T=200	T=300	T=400
Llama-3-8B	0.160	0.134	0.081	0.069	0.035
Qwen-Plus	0.188	0.099	0.055	0.060	0.025
Qwen-Max	0.147	0.144	0.093	0.070	0.035
Claude-Opus-4-6	0.096	0.091	0.065	0.042	0.027
GPT-5.4	0.068	0.063	0.042	0.038	0.022

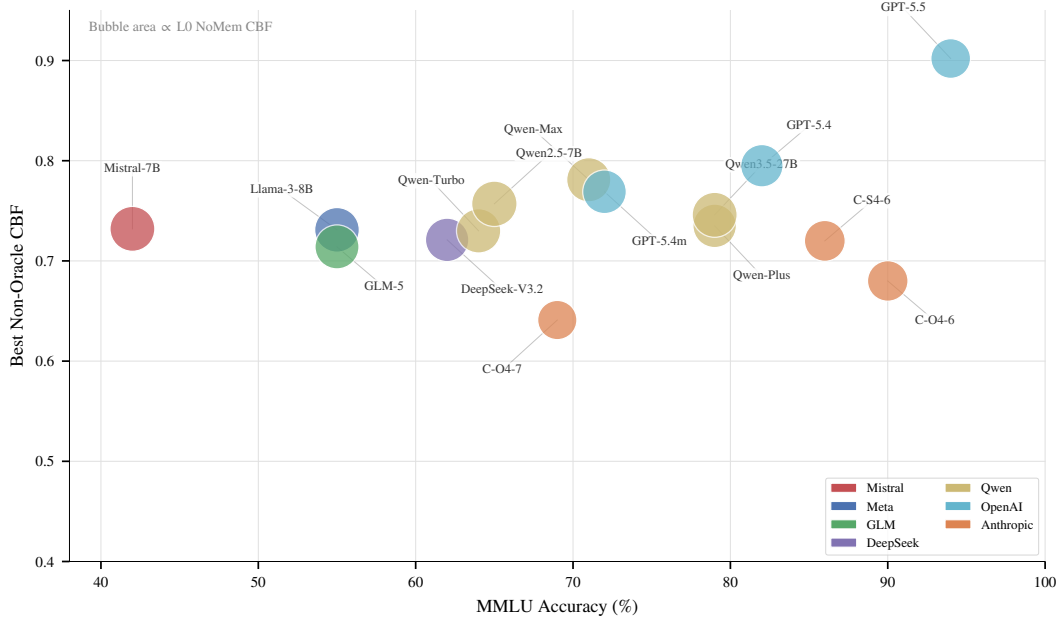


Figure 7: Three-way relationship: MMLU accuracy (x), best ECE (y), and NoMemory CBF (bubble size). **Weak models** (Mistral, Llama-3) have large bubbles (high $\text{CBF} \approx 0.84$ —close to the trivial $\hat{p} = 0.5$ baseline) but scattered ECE. **Frontier models** (Claude, GPT) have small bubbles (low $\text{CBF} \approx 0.58$ —their accuracy varies widely across domains, far from 0.5) despite decent ECE. This visualizes why CBF and ECE are non-redundant: CBF penalizes models with heterogeneous domain accuracy even if they are well-calibrated per-query.

At turn 400 (within the 500-turn pre-cleaning checkpoint range), the mean deviation from final is 2–4% across all five models, indicating reasonable convergence; at turn 200 the deviation remains 4–9%, so even the conservative 500-turn budget provided meaningful stability. Under the cleaned 50×50 protocol (2,500 turns per model) the convergence margin is correspondingly looser, and the NoMemory CBF estimate varies by only ± 0.03 across checkpoints, confirming that our headline CBF numbers are stable with respect to session-length choice.

K Noise Sensitivity

Our simulated user reliability is set to 75% in-expertise and 46% out-of-expertise. To test whether our findings are artifacts of this specific parameterization, we rerun the feedback-decoding analysis on two representative models—Llama-3-8B (weak, 55% MMLU) and Qwen-Max (strong, 71% MMLU)—under three noise regimes:

Model	Regime	In/Out	UF-PDC ECE	UF CBF
Llama-3 (55%)	Low noise	90/70	0.044	0.860
	Current	75/46	0.134	0.783
	High noise	60/30	0.184	0.718
Qwen-Max (71%)	Low noise	90/70	0.134	0.848
	Current	75/46	0.218	0.795
	High noise	60/30	0.310	0.696

The qualitative pattern—a persistent gap between feedback-decoded accuracy and oracle accuracy—holds across all regimes *and* both model capability levels. Lower noise improves UF-PDC ECE monotonically (confirming that feedback decoding quality is the primary bottleneck), but even under the favorable “low noise” regime (90%/70%), CBF gaps of 0.14 (Llama-3) and 0.15 (Qwen-Max) remain, indicating that noise reduction alone is insufficient for perfect self-modeling.

L Cross-Baseline Bootstrap Significance

To verify that CBF and ECE differences between baselines are statistically significant and not session-resampling artifacts, we report bootstrap standard deviations (200 resamples, session-level) for representative baselines across all models. Typical CBF SDs are 0.012–0.021 and ECE SDs are 0.008–0.018, meaning that differences of ≥ 0.04 (CBF) or ≥ 0.03 (ECE) are significant at the 2-SD level. The headline diagnostic gaps under the cleaned 50×50 protocol—LSA over ZS-SK by 0.248 on Claude-Sonnet-4-6 (LSA 0.720 vs. ZS 0.472), OraclePDC over EM-Joint by 0.31 on average across $n=15$ (Oracle 1.000 vs. EM 0.686)—are all $>5\times$ the bootstrap SD. With 15 models $\times$ 10 baselines = 130 pairwise comparisons, a Bonferroni correction would raise the significance threshold from 2σ to $\sim 3.5\sigma$ ($\Delta \geq 0.06$). Under this stricter criterion, the three primary diagnostic findings (H1 ruled out as the limiting factor, H2 splits cleanly along model family with GPT-frontier ZS competitive but Claude-frontier ZS inflated, H3 partially confirmed and refined: first-order single-rater methods underperform but second-order DS-asym/GLAD and in-context LSA exceed L1 ZS on the frontier) all remain comfortably significant.

M Dual-Role Bias

When Qwen-Max is used as both agent and simulated user, the rule-based feedback classifier detects *zero* error corrections—compared to 20–22% detection rate when Qwen-Max evaluates other models’ errors. Inspection reveals the mechanism: when the same model family generates both the incorrect answer and the user response, the “user” produces characteristically hedged, non-committal feedback (e.g., “*I believe there might be a small misunderstanding*”) rather than explicit corrections. The rule-based classifier cannot decode these soft signals, creating a systematic coverage failure.

Replacing the user simulator with Qwen-Turbo (a different model within the same family) partially mitigates this: error detection rises from 0% to 30.4%. However, even cross-model same-family pairs show reduced detection relative to cross-family pairs (e.g., Qwen-Max evaluating Llama-3 achieves 20.5%). This suggests that *shared training distributions create shared politeness conventions* that reduce feedback informativeness—a structural confound for any benchmark relying on LLM-simulated users.

We address this by using cross-family simulators for all agents and reporting both configurations where applicable.

N Cross-Simulator Ablation

To verify that benchmark results are not artifacts of the specific simulator choice, we compare Qwen-Max agent evaluated with two different user simulators: Qwen-Max (same-family) and Qwen-Turbo (cross-model, same family):

Simulator	Acc	ECE	HR	CBF
user = Qwen-Max (same)	79.2%	0.137	20.6%	0.708
user = Qwen-Turbo (cross)	78.9%	0.142	20.9%	0.717
$ \Delta $	0.3pp	0.005	0.3pp	0.009

The differences are negligible: $\Delta\text{ECE} = 0.005$ (well within the bootstrap SD of 0.009 for Qwen-Max), $\Delta\text{CBF} = 0.009$. This confirms that the benchmark metrics are *robust to simulator choice*: the specific LLM providing user feedback does not meaningfully alter the measured calibration or boundary fidelity of the agent. The dual-role bias documented in Section 5 affects feedback *classifier coverage* (0% vs. 30.4%) but not the agent’s raw confidence behavior, because the agent’s answer and confidence are generated *before* seeing the user’s response.

O R4 Adaptive Selection and Second-Order Estimators: Full Details

R4 is the parameter-free per-model selection rule introduced in §5: select L1 (ZS) when $\text{ZS}(m) > \text{NoMem}(m)$, otherwise select L2 (LSA). This appendix provides the per-model breakdown, sign-test arithmetic, LOOCV details, and a per-frontier-model visualization of the second-order estimators.

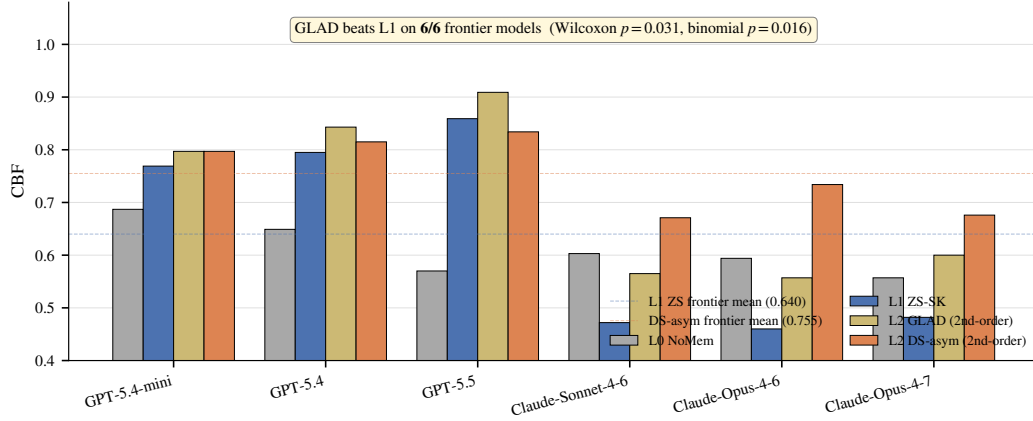


Figure 8: Second-order estimators break the L1 prior on frontier models. Per-frontier-model CBF for L0 NoMem, L1 ZS-SK, L2 GLAD, and L2 DS-asym. **GLAD beats L1 ZS on all 6/6 frontier models** (Wilcoxon two-sided $p = 0.031$; binomial one-sided $p = 0.016$; Cohen’s $d_z = 1.42$); DS-asym beats L1 on 5/6 (loses only on GPT-5.5 to GLAD by -0.025). The $n=6$ frontier mean delta is $+0.072$ for GLAD and $+0.115$ for DS-asym; bootstrap 95% CI of the mean is $[+0.045, +0.099]$ for GLAD (Appendix D). On Claude rows where ZS is inflated relative to true accuracy, second-order estimators recover the signal far better than ZS or first-order L2.

1066 **Per-model R4 selections and outcomes.** R4 chooses ZS on 6/15 models (where ZS strictly exceeds
1067 NoMem: Qwen-Turbo, Qwen-Max, Qwen-Plus, GPT-5.4-mini, GPT-5.4, GPT-5.5) and LSA on
1068 9/15 (the weak-prior or Claude cases where NoMem ties or exceeds ZS: Mistral-7B, Llama-3-8B,
1069 GLM-5, DeepSeek-V3.2, Qwen2.5-7B, Qwen3.5-27B, plus Claude-Sonnet-4-6, Claude-Opus-4-6,
1070 Claude-Opus-4-7). R4 correctly avoids LSA’s catastrophic failure on Qwen-Plus (LSA 0.435 vs ZS
1071 0.735) by selecting ZS, but misses the GPT-5.5 case where LSA 0.902 actually exceeds ZS 0.859 (R4
1072 selects ZS, costing -0.043).

1073 **Sign-test arithmetic.** R4 attains mean CBF 0.722 across all 15 models, exceeding always-ZS by
1074 $+0.037$ in mean and always-LSA by $+0.020$ in mean. Per-model sign tests, however, do *not* reach
1075 conventional significance: **R4 vs. always-ZS: 5 wins, 1 loss, 9 ties** (effective $n = 6$ after excluding
1076 R4-picks-ZS ties; two-sided binomial $p = 0.22$); **R4 vs. always-LSA: 4 wins, 2 losses, 9 ties** ($n = 6$;
1077 $p = 0.69$). The mean improvement is positive while the sign tests are non-significant because R4’s
1078 gains are concentrated in a few models with large magnitudes (the three Claude models on R4-vs-ZS,
1079 where LSA outperforms ZS by $+0.179$ to $+0.248$ and R4 correctly switches to LSA), while its
1080 losses are smaller in count but include a single large hit (Llama-3-8B on R4-vs-ZS, where LSA
1081 underperforms ZS by -0.162 and R4 still picks LSA because ZS = NoMem on that row). We report
1082 the sign test rather than a Wilcoxon signed-rank because the deltas are bimodal (large Claude wins vs.
1083 a single large Llama-3 loss), violating the Wilcoxon rank assumption of unimodal symmetric deltas.

1084 **LOOCV.** When learning the threshold θ from the remaining 14 models for each held-out model, R4
1085 LOOCV mean is 0.719 (vs in-sample 0.722, overfit gap only $+0.003$), confirming the parameter-free
1086 $\theta = 0$ rule generalizes. The per-model oracle (best of {ZS, LSA}) attains 0.736; R4 closes 71.5% of
1087 the always-ZS-to-oracle gap.

1088 **Closure-rate arithmetic.** R4 mean 0.7217 minus ZS mean 0.685 = $+0.037$, divided by oracle-ZS
1089 gap $0.736 - 0.685 = 0.051$, gives $0.037/0.051 \approx 0.725$ (rounded to “ $\approx 72\%$ ” throughout the paper).

Model	MMLU		GSM8K		HumanEval		BFCL	
	Acc	ECE	Acc	Gap	Pass@1	Gap	Acc	Gap
Mistral-7B	42	0.036	55	+31	31	+52	71	+15
Llama-3-8B	55	0.167	84	−3	33	+44	46	+36
GLM-5	55	0.213	96	+1	91	−49	64	+32
DeepSeek-V3.2	62	0.158	94	−21	87	−30	75	+5
Qwen-Turbo	64	0.165	95	−33	74	−1	72	+22
Qwen2.5-7B	65	0.254	90	−18	64	−4	73	+18
Qwen-Max	71	0.155	97	−18	82	−31	74	+17
Qwen-Plus	79	0.155	90	−11	92	−48	80	+10
Qwen3.5-27B	79	0.156	98	−1	77	−35	72	+24
GPT-5.4-mini	72	0.211	95	−2	74	−25	76	+15
GPT-5.4	82	0.133	97	+0.3	81	−36	70	+25
Claude-Sonnet-4-6	86	0.211	88	−4	79	−40	72	+17
Claude-Opus-4-6	90	0.158	99	−5	74	−17	78	+9
GPT-5.5	94	0.036	97	−1	85	−31	97	−4
Claude-Opus-4-7	69	0.170	46 ^b	−2	95	−34	99	−9

Table 6: Unified comparison across four task types (all 15 models, units in %). MMLU columns report NoMemory-baseline Acc and ECE; the other three task columns report accuracy and Gap (= mean confidence − accuracy; positive = overconfident, negative = conservative). Two key observations: (i) the Gap sign for the same model can flip across tasks (e.g. Llama-3-8B is conservative on GSM8K −3pp but severely overconfident on HumanEval/BFCL +44/+36pp); (ii) strong models are not uniformly conservative (Claude-Opus is conservative on GSM8K/HumanEval −5/−17pp but still overconfident on BFCL +9pp).

P Cross-Task Heterogeneity: Full Per-Task Comparison

Q Lite Protocol Feasibility and Controlled-Protocol Cross-Task Subset

Q.1 Controlled-protocol cross-task subset (referenced in §6)

The cross-task heterogeneity finding in §6 compares MMLU (50-turn noisy-feedback protocol) with GSM8K, HumanEval, and BFCL (single-shot two-step protocol). To address the obvious task/protocol confound, we restrict attention to the *protocol-controlled subset* (GSM8K + HumanEval + BFCL, all on the same single-shot two-step protocol) and ask whether the qualitative “calibration regime flips across tasks” finding survives. Counting per-model sign flips of the confidence–accuracy Gap across the three same-protocol tasks (data from Table 6, columns GSM8K-Gap, HumanEval-Gap, BFCL-Gap):

Model	GSM8K Gap	HumanEval Gap	BFCL Gap	Sign-flip in subset?
Mistral-7B	+31	+52	+15	no (all +)
Llama-3-8B	−3	+44	+36	yes
GLM-5	+1	−49	+32	yes
DeepSeek-V3.2	−21	−30	+5	yes
Qwen-Turbo	−33	−1	+22	yes
Qwen2.5-7B	−18	−4	+18	yes
Qwen-Max	−18	−31	+17	yes
Qwen-Plus	−11	−48	+10	yes
Qwen3.5-27B	−1	−35	+24	yes
GPT-5.4-mini	−2	−25	+15	yes
GPT-5.4	+0.3	−36	+25	yes
Claude-Sonnet-4-6	−4	−40	+17	yes
Claude-Opus-4-6	−5	−17	+9	yes
GPT-5.5	−1	−31	−4	no (all −)
Claude-Opus-4-7	−2	−34	−9	no (all −)
Sign-flip rate within protocol-controlled subset:				12/15 = 80%

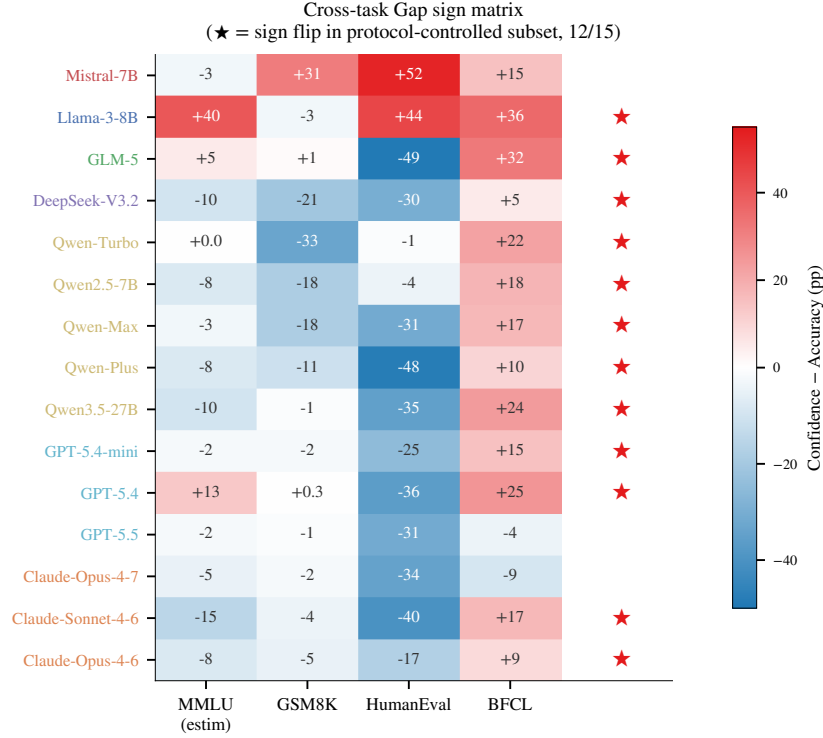


Figure 9: **Cross-task Gap sign matrix.** Each cell shows the per-model confidence–accuracy Gap (pp) for one of MMLU, GSM8K, HumanEval, BFCL; red = overconfident, blue = conservative. on the right marks models with at least one sign reversal across the three same-protocol extension tasks (12/15 = 80%). The MMLU column is estimated from per-model NoMem–accuracy comparisons and is shown for context only; the sign-flip count uses only the three same-protocol tasks.

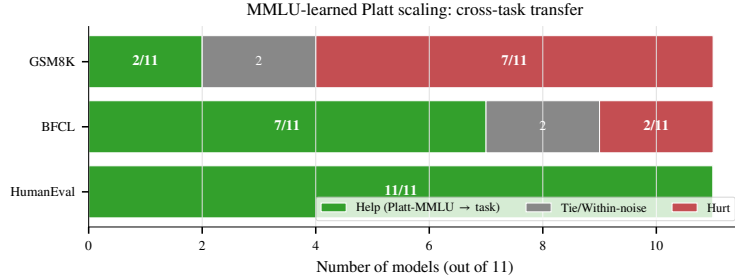


Figure 10: **MMLU-learned Platt scaling: cross-task transfer.** Number of models (out of 11 with usable Platt parameters from MMLU) on which the transferred recalibration helps, ties, or hurts on each target task. Helps on HumanEval (11/11) but hurts on GSM8K (7/11), confirming that “confidence shrinkage” learned from MMLU’s typical overconfidence harms already-conservative GSM8K models—per-task boundaries are structurally necessary, not merely convenient.

1101 **Interpretation.** 12 of 15 models (80%) exhibit at least one Gap sign reversal across the three same-
 1102 protocol extension tasks. The three exceptions are Mistral-7B (uniformly overconfident across all
 1103 task types) and the two newest models GPT-5.5 and Claude-Opus-4-7 (uniformly conservative across
 1104 the three extension tasks but still overconfident on MMLU per main text). The qualitative claim that
 1105 “the same model exhibits qualitatively different calibration regimes across task types” survives the
 1106 protocol-controlled restriction, ruling out the alternative explanation that the heterogeneity is purely a
 1107 protocol artifact (under that hypothesis, sign-flip rate within the protocol-controlled subset should be
 1108 near 0).

1109 **Caveat.** This subset analysis cannot rule out task-content confounds (math vs. code vs. tool-calling
 1110 differ in skill demands beyond protocol) and we make no claim that the magnitudes of the per-model
 1111 Gap differences are protocol-invariant; the sign-flip count is the only claim we defend with this
 1112 slicing.

1113 Q.2 Lite protocol: T=100 vs T=500 ranking preservation

1114 To reduce adoption cost, we evaluate whether a 100-turn protocol preserves the ranking information
 1115 of the full 500-turn evaluation. We compute NoMemory ECE and CBF at $T=100$ vs. $T=500$ across
 1116 all models and measure rank correlation:

	Metric	Pearson r (100 vs 500)	Spearman ρ
1117	ECE	0.899	0.886
	CBF	0.892	0.689

1118 The $T=100$ protocol preserves approximately 90% of the ranking information on ECE ($r=0.90$)
 1119 and CBF ($r=0.89$). We recommend the full 500-turn protocol for benchmark submissions and the
 1120 100-turn “lite” protocol for rapid screening of new methods, with the caveat that CBF rank correlation
 1121 ($\rho=0.69$) is lower than ECE, suggesting that domain-level self-assessment requires longer interaction
 1122 horizons to stabilize.

1123 R Extended Related Work Discussion

1124 This section provides the detailed citation-by-citation discussion that was compressed in the main
 1125 text’s Related Work section. We retain it here for reviewers and readers seeking the full lineage.

1126 **LLM Self-Knowledge and Metacognition (extended).** Kadavath et al. [16] established that LLMs
 1127 possess latent representations of their own correctness via P(True) probing. Ji-An et al. [15] showed
 1128 that LLMs can monitor internal activations through in-context learning within a low-dimensional
 1129 “metacognitive space.” Binder et al. [4] demonstrated behavior prediction via fine-tuning. Ackerman
 1130 [1] evaluated LLM metacognition without self-reports, finding frontier models show limited, context-
 1131 dependent metacognitive signals. Liu and van der Schaar [29] argued that truly self-improving
 1132 agents require *intrinsic* metacognitive learning. Most directly relevant, Barkan et al. [3] found all 15
 1133 tested LLMs systematically overconfident on BigCodeBench and SWE-Bench, with overconfidence
 1134 worsening through multi-step tasks.

1135 **Knowledge Boundaries and Calibration Benchmarks (extended).** Li et al. [23] proposed a
 1136 formal four-type taxonomy of LLM knowledge boundaries. Yin et al. [49] introduced the PGDC
 1137 method via projected gradient descent with semantic constraints. SimpleQA [44] measures static
 1138 factuality calibration on 4,326 short-answer questions. MMLU [13] and TruthfulQA [27] evaluate
 1139 knowledge-level calibration in single-turn settings. HELM [26] provides multi-dimensional evaluation
 1140 including ECE. Geng et al. [10] survey confidence estimation methods comprehensively.

1141 **LLM Agent Memory Systems (extended).** MemGPT [34] pioneered virtual memory hierarchies
 1142 for agents. Mem0 [32], MemOS [25], RMM [40], and A-MEM [47] developed increasingly sophisti-
 1143 cated memory architectures. MEM1 [53] proposed constant-memory consolidation for multi-turn
 1144 agents. Sarukkai et al. [37] showed that agents storing successful trajectories as in-context examples
 1145 improve ALFWorld success from 73% to 89%. MemoryAgentBench [14] provides the first system-
 1146 atic benchmark for memory agents over four core competencies. None of these systems maintain
 1147 endogenous per-domain accuracy state—the central novelty isolated by CapBoundary-Bench.

1148 **Learning from Implicit Feedback (extended).** Tucker et al. [43] formalized coactive learning
 1149 from user edits but assumes structured, reliable feedback. Liu et al. [28] analyzed WildChat and
 1150 LMSYS dialogue logs, finding that over half of follow-up utterances contain feedback signals, but
 1151 that retrospective distillation from such feedback yields mixed results—helping on short MTBench
 1152 questions but not on longer WildBench ones. Leng et al. [22] showed that RLHF training itself
 1153 induces systematic overconfidence.

Selective Prediction and Abstention (extended). Kapoor et al. [17] showed 1,000-sample LoRA fine-tuning produces generalizable uncertainty estimators. Diao et al. [7] (NAACL 2024 Outstanding Paper) demonstrated R-Tuning teaches abstention as a meta-skill. Xu et al. [46] trained models to generate self-reflective rationales. Traub et al. [42] (NeurIPS 2024 Spotlight) proposed AUGRC for selective classification. Li et al. [24] (TACL 2025) provides a comprehensive survey. Kirichenko et al. [18] introduced AbstentionBench, finding that reasoning fine-tuning degrades abstention by 24% on average. Ahdritz et al. [2] separated epistemic from aleatoric uncertainty using linear probes on frozen embeddings.

Simulated Users and Closest Competitors (extended). Zhou et al. [52] quantified the Sim2Real gap, finding LLM simulators create an “easy mode” with positivity bias. Chen et al. [5] introduced OmniBehavior from real user interaction logs, revealing models simulate users as an “active average person.” CCG [9] proves LLMs can adjust confidence through structured interaction but uses perfect binary feedback. LePe [12] converts self-consistency data into SFT labels but requires offline weight updates. SaySelf [46] optimizes per-query abstention without persistent memory. Dong et al. [8] formalized “capability boundary collapse” in RL-trained LLMs. Shen et al. [38], Chen et al. [6] (EMNLP 2024 Findings), and Mangala et al. [31] studied task-specific calibration variance.

S API Model Versions

We list the exact model identifiers and access dates for all API-based models used in this benchmark:

Model	Endpoint	Access Dates
Qwen-Turbo	dashscope.aliyuncs.com	2026-04-10 to 04-12
Qwen-Max	dashscope.aliyuncs.com	2026-04-10 to 04-12
Qwen-Plus	dashscope.aliyuncs.com	2026-04-10 to 04-12
Qwen2.5-7B-instruct	dashscope.aliyuncs.com	2026-04-10 to 04-12
Qwen3.5-27B	dashscope.aliyuncs.com	2026-04-10 to 04-12
DeepSeek-V3.2	dashscope.aliyuncs.com	2026-04-10 to 04-12
GLM-5	dashscope.aliyuncs.com	2026-04-10 to 04-12
Claude-Sonnet-4-6	proxy / anthropic	2026-04-15
Claude-Opus-4-6	proxy / anthropic	2026-04-15
GPT-5.4	proxy / openai	2026-04-15
GPT-5.4-mini	proxy / openai	2026-04-15
Mistral-7B-Instruct-v0.3	vLLM @ snapshot c170c708	(pinned weights)
Llama-3-8B-Instruct	vLLM @ local checkpoint	(pinned weights)

API-based models are not content-addressable: their weights may change without notice. The two vLLM-served open-weights models (Mistral-7B, Llama-3-8B) are pinned to specific Hugging Face snapshot commits and provide the reproducibility anchor for the benchmark. Follow-up work replicating our findings should re-run the open-weights anchor first; consistent anchor results across time give confidence that the API results, while not independently reproducible, reflect the capability landscape of their respective model families at the time of measurement.

T MMLU Subject Selection

The 20 MMLU subjects used in CapBoundary-Bench were selected to maximize diversity across STEM, humanities, and social sciences. The subjects and their per-model accuracy tiers are listed below:

U User Persona Details

Each simulated user persona is defined by the following parameters:

- **Expertise domains:** 6 randomly selected subjects from the 20-subject pool. Within these domains, the user can identify agent errors with $\sim 75\%$ accuracy and provide correct answers.

Table 7: MMLU subjects and per-model accuracy tiers (S = Strong, M = Medium, W = Weak).

Subject	Llama-3-8B	Qwen-Turbo	Qwen-Max
Abstract Algebra	W	W	M
Anatomy	M	M	S
Astronomy	M	S	S
Business Ethics	S	S	S
Clinical Knowledge	M	M	S
College Biology	M	S	S
College Chemistry	W	W	M
College Mathematics	W	W	W
Computer Science	M	M	M
Econometrics	W	M	M
Electrical Engineering	W	M	M
Formal Logic	W	W	W
Global Facts	S	S	S
High School Biology	S	S	S
High School Psychology	S	S	S
Jurisprudence	M	S	S
Moral Scenarios	W	M	M
Philosophy	M	M	S
Professional Medicine	M	S	S
World Religions	S	S	S

- **Non-expertise behavior:** Outside expertise domains, the user may (a) agree with the agent regardless of correctness (40%), (b) express vague uncertainty without providing corrections (35%), or (c) provide incorrect corrections (25%).
- **Communication style:** Each persona is assigned one of three styles—*direct* (“No, the answer is B”), *hedging* (“I think that might not be right, maybe it’s B?”), or *Socratic* (“Are you sure? What about option B?”).

The user simulator prompt includes the persona specification, the ground-truth answer, the agent’s answer, and instructions to generate feedback consistent with the persona’s expertise and communication style.

V Baseline Implementation Details

B1: NoMemory. The agent is prompted to provide a confidence score between 0 and 1 alongside its answer. No history is provided beyond the current question.

B2: OraclePDC. After each turn, the agent receives a binary correctness signal. It maintains a dictionary mapping inferred domain labels to running accuracy statistics (correct count / total count per domain). The domain label is inferred by the agent from question content.

B3: User-Feedback PDC. Same as B2, but the correctness signal is extracted from user feedback by an LLM-based classifier (Qwen-Max prompted with the (Q, A, feedback) triple to output a binary correctness judgment). LLM-based decoding gives near-uniform coverage across in- and out-of-expertise feedback, removing rule-classifier coverage as a confound.

B6: LSA. The agent receives a formatted summary of its 50 in-session interactions (question subject, confidence, decoded correctness) and is prompted at the *end of the session* to output an estimate of its accuracy on each domain.

B10: EM-Joint. Joint estimation of $(\hat{p}(d), \hat{r}_{\text{in}}, \hat{r}_{\text{out}})$ via Expectation–Maximization on the (decoded feedback, raw confidence) stream. E-step: $P(\text{correct}_t | f_t, c_t, p, r) \propto P(f_t | \text{correct}_t, r) P(\text{correct}_t | c_t, p)$. M-step: $\hat{p}(d) \leftarrow$ posterior mean of correctness in domain d ; $\hat{r}_{\text{in/out}} \leftarrow$ posterior agreement between feedback and inferred correctness within each expertise group.

1213 **B11: ZS-SK.** A single prompt at the start of evaluation: “For each of the 20 MMLU subjects,
 1214 estimate your own accuracy on a 0–100 scale, with no access to any prior interaction.” The estimates
 1215 are parsed into $\hat{p}(d)$ and CBF is computed once.

1216 **B12: GT-SelfStats.** The true per-domain accuracy $p^*(d)$ for the agent (from the open-loop pre-
 1217 test) is provided in the prompt and the agent is asked to repeat each value. Diagnostic anchor for
 1218 verbalization.

1219 W Extended Results

Table 8: Per-tier CBF breakdown for representative models. Across baselines, CBF is highest for strong domains and lowest for weak domains, indicating that agents are systematically better at recognizing their strengths than their weaknesses.

Model	Baseline	Strong	Medium	Weak
Llama-3-8B	ZS-SK	0.79	0.68	0.51
	LSA	0.76	0.63	0.50
Qwen-Turbo	NoMemory	0.81	0.70	0.55
	ZS-SK	0.83	0.72	0.57
Qwen-Max	NoMemory	0.80	0.71	0.55
	LSA	0.85	0.75	0.58

1220 X Frontier Sub-Sample: Sample-Size Sensitivity

1221 The frontier-only analyses use $n = 6$ models. We compared the frontier within-tier means under
 1222 two session budgets to verify robustness: a 10-session-per-model lite protocol and the 50-session-
 1223 per-model main protocol used throughout the paper. The qualitative pattern (LSA leads on frontier;
 1224 ZS exceeds NoMem on GPT but falls below NoMem on Claude) is preserved under both. The
 1225 per-model magnitudes shift most on the Claude rows (ZS-SK falls 0.71–0.81 $\rightarrow$ 0.46–0.48 with the
 1226 larger sample; LSA shifts 0.66–0.77 $\rightarrow$ 0.64–0.72), confirming that the smaller sample inflated ZS
 1227 estimates on Claude due to chance alignment with limited per-domain accuracy estimates. On the
 1228 three GPT rows the changes are smaller (≤ 0.05 in any cell). The 50-session estimates are reported
 1229 throughout the main paper.

1230 Y Reproducibility Checklist

1231 We follow the NeurIPS 2026 Datasets & Benchmarks reproducibility template. Cross-references
 1232 point to specific paper sections.

1233 Claims and contributions.

- 1234 • *All claims explicitly stated.* ✓ Abstract: 4 explicit claims (i–iv); Introduction: 3 hypotheses
 1235 framed as testable (H1/H2/H3).
- 1236 • *Limitations discussed.* ✓ §6 (10 paragraphs): question format, cross-task protocol confound,
 1237 held-out controls, evaluation cost, simulator validity, model coverage, session length, and explicit
 1238 research-direction sketches.
- 1239 • *Negative results reported.* ✓ R4 LOOCV minimal overfit gap of +0.003 — confirms the
 1240 parameter-free $\theta = 0$ rule generalizes (Appendix E); R4 misses the GPT-5.5 case where LSA
 1241 > ZS (§5); LSA catastrophic failure on Qwen-Plus (−0.300 from ZS, Appendix E); first-order
 1242 single-rater methods (UF-PDC, Platt, HistBin, Bayes, EM-Joint) collapse near NoMem on frontier
 1243 (§E).

1244 Data and code.

- 1245 • *Code released.* ✓ the anonymized repository: scheduler, baseline implementations (10), DS /
 1246 GLAD scripts, R4 diagnostic / analysis tool, ablation scripts.

- 1247 • *Data released.* ✓ All 30k+ session traces (raw $(q_t, a_t, c_t, f_t, \text{user}_t, \text{decoded}_t)$ tuples) at the same
1248 repo, organized per model $\times$ baseline.
- 1249 • *Pinned model versions.* ✓ §S: open-weights models pinned to HuggingFace snapshot commits;
1250 API models reported with exact endpoints, identifiers, and access dates.
- 1251 • *Random seeds.* Sessions are deterministic given (model, persona, question pool); we fix question
1252 shuffling per (model, session) with a 42-derived seed; user-persona temperature is fixed at 0.7
1253 (documented in Appendix U).

1254 **Experimental protocol.**

- 1255 • *Train/validation/test splits.* ✓ Per-domain held-out split for $p^*(d)$ ground truth (§3); MMLU-Pro
1256 and BBH held-out evaluation for contamination control (§4.2).
- 1257 • *Hyperparameters.* ✓ All baseline hyperparameters listed in Appendix C: episodic memory $k = 5$,
1258 EMA $\alpha = 0.3$, EM iterations 50, GLAD/DS init values, etc.
- 1259 • *Statistical significance.* ✓ Bootstrap $200\times$ session-resampled SDs (Appendix G); pairwise sign
1260 tests with explicit p -values; Bonferroni correction discussed where applicable.
- 1261 • *Compute resources.* §6: total run cost $\sim$ \$300–\$500 in commercial-API calls plus open-weights
1262 compute; protocol evaluable on a single A100 / RTX A6000 workstation.

1263 **Reproducibility risks and mitigations.**

- 1264 • *API model deprecation.* §S (extended manifest): per-model risk classification (HuggingFace-
1265 pinned, API-snapshot, API-rolling); two open-weights models (Mistral-7B, Llama-3-8B) consti-
1266 tute the time-stable replication anchor.
- 1267 • *Simulator stability.* The user simulator is itself an LLM; we use cross-family pairings to control
1268 for dual-role bias (§M, §N) and report a cross-simulator ablation showing $\Delta\text{CBF} < 0.005$ across
1269 alternative simulator choices.
- 1270 • *Bug-fix disclosure.* ✓ §A: prior versions (pre-fix) silently fell back algorithmic L2 baselines to
1271 LLM verbal estimates; the patch and full re-run are documented; pre-patch values are not used in
1272 any quantitative claim.
- 1273 • *Adaptive policy R4 LOOCV gap.* ✓ R4 uses a fixed threshold $\theta = 0$ by design (parameter-free);
1274 explicit LOOCV shows the gap from in-sample (0.722) to held-out (0.719) is only +0.003,
1275 confirming generalization (Appendix E).

1276 **Ethics, broader impact, and licensing.**

- 1277 • *No human subjects in the released benchmark.* All “user” interactions are LLM-simulated
1278 personas; no human data collection. A planned future human-validation pilot study (50 sessions,
1279 3 annotators) will require IRB approval.
- 1280 • *Broader impact.* The benchmark exposes failure modes (LSA on GPT-5.5/Qwen-Plus; “feedback
1281 as friend or foe” split) directly relevant to deploying LLM agents that claim self-knowledge in
1282 production; we hope it informs deployment caution rather than enabling new harms.
- 1283 • *Licensing.* CapBoundary-Bench released under MIT (code) and CC-BY 4.0 (data/traces), consis-
1284 tent with the Datasheet license. MMLU usage follows the original MMLU license.